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MA Thesis

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Veri(dis)similitude: The Rhetorical Force Generated by Virtual Reality Technology

A Note on Quotations and Abbreviations

Due to the rather innovative and unique way in which Heidegger employs punctuation and other stylistic features, any quotation from Heidegger used in this paper remains unchanged. Thus, everything within the first set of quotation marks appears exactly as it does in the text from which it is quoted. Accordingly, please excuse the few instances where two sets of quotation marks appear next to each other.

The following abbreviations for titles are used:

<i>BT</i>	<i>Being and Time</i>
<i>GoM</i>	<i>A Grammar of Motives</i>
<i>Introduction</i>	<i>Introduction to Metaphysics</i>
ISPR	International Society for Presence Research
<i>LSA</i>	<i>Language as Symbolic Action</i>
<i>PC</i>	<i>Permanence and Change: An Anatomy of Purpose</i>
QCT	“The Question Concerning Technology”
<i>RoM</i>	<i>A Rhetoric of Motives</i>

Veri(dis)similitude: The Rhetorical Force Generated by Virtual Reality Technology

In what follows we shall be *questioning* concerning technology. Questioning builds a way. We would be advised, therefore, above all to pay heed to the way, and not to fix our attention on isolated sentences and topics. The way is a way of thinking. All ways of thinking, more or less perceptibly, lead through language in a manner that is extraordinary. (Heidegger, QCT 3)

My project is not so ambitious as Heidegger's; for I do not seek to define the essence of technology as a whole, but rather the unique rhetorical force generated by one specific technology: Virtual Reality (VR). Therefore, in what follows we shall be *questioning* concerning virtual reality technology. Virtual reality, a term purportedly coined in 1987—or at least popularized—by computer scientist, author and composer Jaron Lanier, has become a trending topic among computer scientists, game and software designers within the last few years (Lowood). Such systems use computer models and simulations that enable the user to dive into an interactive simulation. These computer-generated worlds or virtual environments (VEs) immerse the user in a simulated reality through the use of interactive devices such as Head Mounted Displays (HMDs), controllers, and data gloves. In the most common VR format the user dons an HMD through which they stereoscopically view the images of a simulated space and which tracks the user's movement allowing them to “physically” navigate that virtual space. Thus, VR technology functions partly through sensory deprivation for the HMD cuts the user off from both visual and auditory sense-data, blocking out our sensory perception of the real world and helping to establish virtual reality in its place.

We shall be questioning concerning *virtual reality technology*; more specifically, we shall be questioning concerning the unique rhetorical force generated by VR technology. Questioning builds a way and this way is a way of thinking. But any question or inquiry, as Heidegger argues,

“is a seeking [Suchen]” and “[e]very seeking gets guided beforehand by what is sought. Inquiry is a cognizant seeking for an entity both with regard to the fact that it is and with regard to its Being as it is” (BT 24). Thus, to work out the question of the rhetorical force generated by VR technology—with regard to the fact that it is and to its Being as it is—we must make VR technology transparent as an entity, the distinctive features of which “will come fully to light only when we have delimited it adequately with regard to its function, its aim, and its motives” (29).

What is the unique rhetorical force generated by VR technology? This way of questioning is directed beforehand by what is sought; in other words, this questioning is guided by a motive. Up until a very unique event I experienced within the virtual program *Gnomes & Goblins*, this questioning was motivated chiefly by the lack of current rhetorical and phenomenological investigations of a specific virtual reality program. *Gnomes & Goblins* is comedian, actor, and director Jon Favreau’s experiment with virtual reality. Blurring the boundaries between movie and game, *Gnomes & Goblins* transports the user to a magical, enchanted forest. Free to wander and explore the hollowed-out treehouses as well as interact with the environment itself, the user must build a relationship with one of the small and timid goblins who lives there. Over the course of my research I have observed numerous instances where the user reacts to the virtual world as if it were “real”; it is not uncommon, for instance, to see a user attempt to lean against a tree within the VE only to stumble when their hand passes right through the virtual object. Such an occurrence can, and I will discuss this in depth later on, largely be attributed to the program moving the user to a state of, what is commonly known as, “presence”: a sense that they are, to some extent, psychologically and physically present within the VE.

This project, however, was catalyzed by a heretofore unique experience I had while analyzing the rhetorical design features underlying the VE of the program *Gnomes & Goblins* which are intended to induce a state of presence. What I discovered is that there are various degrees of presence. After exploring this enchanted forest for a period of time in which I, somewhat subconsciously, completed the tasks necessary to trigger the next narrative sequence, I was gifted a magical bell by one of the small goblins with whom I had been interacting. Upon ringing the bell, I was dramatically miniaturized down to the same size as the goblin. This transformation produced such a profound vertigo-like sensation that I physically stumbled and was momentarily lost, suffering a sense of dislocation within the virtual world.

Because of the high visual fidelity of certain VEs, it is relatively easy to fall into the common ways through which we understand and interact with objective reality; thus, “leaning” against a virtual tree or “dodging” a virtual object which speeds towards us, actions which are almost instinctive responses, do not require as much work on the part of the program to illicit such actions. We will instinctively duck to avoid, say, a storm trooper’s blaster fire, knowing throughout the entire experience that the laser headed our way is not real. Even though the user may not have entered into a state of true presence, because the virtual *Star Wars* world appears familiar enough to our expectations of such a ‘world,’ and the mechanics for acting and interacting within that ‘world’ are similar to those of objective reality, the program is able to illicit an unconscious physical response despite us knowing full well that it is nothing more than a game. The degree of presence necessary to produce such an effect resulting from the transformation, though, is of a much stronger and slightly different nature. It suggests the possibility that the program is able to evoke a change in the user’s mode of perception so that they understand their human existence as directly and actively implicated within the virtual

world. Thus, to terminologically reformulate the effect of the transformation, this experiential event produced a profound phenomenological shock resulting in a strong sense of physical disorientation—my feeling of vertigo—and an ontological disruption, a momentary disruption in, what Heidegger phrases, my basic state of “Being-in-the-world”: one feature of which he calls “thrownness.” Dasein “is thrown in such a way that, as Being-in-the-world, it is its “there”” (174).

Accordingly, the basic state of presence I shall refer to as, to borrow Heidegger’s term, “thrownness.” This is not a complete state of presence for it is entirely possible to be consciously aware of the HMD on your face and the controllers in your hand and still physically react to an object shot at you. Rather, to be “thrown” into a VE is to be “thrown” into an environment which *matters to*—has meaning for—the user; it is a mode of Being-in a ‘world’ that makes itself intelligible to the user as a ‘world’ which offers possibilities for action and possibilities for being acted upon. To be “thrown” into a VE, then, requires that the specific VE be conceptually and mechanically viable: it must be *familiar* to how we conceive other environments and operate within them. Thus, if a user is placed in an environment which is entirely unfamiliar (an environment, say, composed of nothing but abstract shapes and colors with no visually defined objects or spaces) or if that environment’s mechanics do not cohesively translate (say, if a physical step forward is translated as a relocation of the user’s virtual body, inverting them upside down), then that ‘world’ will have no meaning, it will not *matter to* the user as it will not be intelligible to them.

However, to successfully move a user to a state of true presence requires more: it must, firstly, “throw” the user into a ‘world’ that makes sense and *matters to* them; secondly, it must then condition the user’s behavior and actions within the VE so that they familiarize themselves

with the ‘world’ and the mechanics for operating within it; and, lastly, through that conditioning, psychologically immerse the user, directing their attention away from their existence in objective reality and toward the objects and entities within the VE.¹ In other words, by being “thrown” into a ‘world’ that *matters* to them and encouraged to actively engage with the objects and entities which populate the VE as they so choose, what Heidegger calls our “circumspective dealings” (111), the user becomes “concernfully absorbed” in their activity (146). Thus, I believe psychological immersion can be usefully understood as Heidegger’s “concernful absorption.”

On the other hand, a true state of presence suggests a mode of perception wherein the user has no conscious awareness of the physical equipment (the PC, HMD, and handheld controllers) which generate and mediate the experience, and no conscious understanding of the subject-object dichotomy: no awareness of themselves as a *physical* subject contrasted with a ‘world’ of *virtual* “things” and objects. It is a mode of Being-in a virtual world wherein the user has no conscious awareness of their actual existence in objective reality, a mode where they are their “there” in the virtual world. So then, in light of this, to move a user to a state of true presence in the VE into which they are “thrown” the program must, by directing or conditioning our “circumspective dealings,” “concernfully absorb” the user “in whatever work-world likes closest to” them so that their attention is more focused on the objects and entities within the virtual world than the physical equipment or their existence in objective reality (101).

Accordingly, I shall call this state of true presence, to adapt Heidegger’s phrase, Being-in-a-virtual-world-in-*the*-world. I must pause and note that I differentiate between the common mode of presence and that of true presence or Being-in-a-virtual-world-in-*the*-world because the

¹ According to the International Society for Presence Research (ISPR), “psychological immersion” occurs “when part or all of a person’s perception is directed toward objects, events, and/or people created *by* the technology, and away from objects, events, and/or people in the physical world” (par. 7d).

former does not have enough substance to explain the cause for the disorientation produced by the transformation. To be only “thrown” into a VE the user is its “there” but an awareness of the physical equipment and the user’s existence in objective reality exists to some extent; though the ‘world’ may illicit instinctual responses, it is unable to move the user to a mode of Being-in the VE where the user understands their human existence as actively implicated within the VE. Such a mode of Being is achieved only when the ‘world’ moves the user to a mode of perception where that ‘world’ and their experiences within it are disclosed to them as a formulation of, and I will discuss this in more detail at a later point, what Heidegger calls “thrownness,” “projection,” and “fallenness.” Thus, when a virtual world is able to mimic our “thrownness,” “projection,” and “fallenness” in objective reality it will be able to successfully move the user to a mode of Being-in-*a-virtual-world-in-the-world*.

Once the user has been moved to Being-in-*a-virtual-world-in-the-world*, any disruption in their Being-in-*a-virtual-world-in-the-world*, like the transformation in *Gnomes & Goblins*, produces a mood of “anxiety”: it discloses to us the experience of our Being-in the *Gnomes & Goblins* ‘world’—our Being-as-there—as “anxiety”, a feeling which defamiliarizes or disenchant us from our familiar phenomenological and ontological understanding of our Being-in-*a-virtual-world-in-the-world*. But in such a disruption and the “anxiety” produced “there lies the possibility of a disclosure which is quite distinctive: for “anxiety” individualizes. This individualization brings Dasein back from its falling, and makes manifest to it that authenticity and inauthenticity are possibilities of its Being” (235). It reveals or unconceals not only the sheer rhetorical force generated by an artifact which can mimic our Being-in-*the-world* (of objective reality) and how easily we succumb to a “fallen” state of “inauthenticity” (how easily we enter a mode of perception or mode of Being-in in which we uncritically accept the processes of

perception and understanding through which we conceive reality), but also, and more importantly, this revelation unconceals the rhetorical force generated by VR technology not to be *verisimilitude* (the appearance of being true or real), but rather, what I will call, *veri(dis)similitude*. I use this term because I believe it more precisely captures the unique rhetorical power inherent in virtual reality technology. Indeed, in order to “throw” a user into a virtual world that ‘world’ and the objects and entities which populate it along with the mechanics of operating within it must be *verisimilar* to those of reality (a physical step forward must be translated as a step forward within the VE); but the very allure of VR technology which enables us to become so “concernfully absorbed”—despite the unremovable presence of the hardware which generate and mediate the experience—that we view the VE as *matter[ing] to*—having meaning for—us is its ability to create or manifest rhetorical ‘worlds’ within which the user is located. ‘Worlds’ which, at least in the case of *Gnomes & Goblins*, are *not* limited to earthly laws or places: ‘worlds’ which are *verisimilar* to how we conceive of and interact with reality to the extent that they are *familiar* enough to our normal everyday mode of interacting and understanding objective reality, but which are *dissimilar* to what is familiarly experienced in everyday objective reality. For a virtual world to move a user to a mode of Being-in-a-virtual-world-in-the-world, therefore, it must be *veri(dis)similar* to objective reality and our experiences within it.

As a human made technology, the current, commercially available version of which is marketed, arguably, as a product intended to gratify our desire for entertainment and/or distraction, the structural relations underlying and which govern our experience of a VE should, therefore, be observable entities which can be articulated. For “[n]othing is more imperiously *there* for observation and study,” or so Kenneth Burke argues, “than the tactics people employ

when they would injure or gratify one another” (*GoM* 57; my emphasis). In what follows, I will begin my questioning of the rhetorical force generated by VR technology by first discussing its *technical function* and *rhetorical function*, its *aim*, and its *motive*. I will then move to examine how these elements are made manifest in my experience within *Gnomes & Goblins*—in the features of the VE which work to “throw” the user into a virtual world and, then, move them to a mode of Being-in-a-virtual-world-in-*the*-world. My reasoning behind choosing this artifact, at least initially, is based on three characteristics: firstly, it features a high level of visual fidelity and aesthetic beauty; secondly, unlike some other VR programs, which were originally built for normal PC or console use and thereafter translated into a VR program, *Gnomes & Goblins* was designed and built from the ground up for VR; and, thirdly, it is a playable advertisement or preview of a program soon to be released. In this sense, rather than featuring a trailer which functions to advertise the program, this playable demo—which has increasingly come to replace traditional game trailers—lets the user experience the first 10-15 minutes of actual gameplay. Because this is meant to entice potential consumers to purchase the full version, the preview is designed to have a powerful effect on the user while simultaneously introducing them to the narrative and mechanics of game play. After, I will move to analyze the human experience or mode of encountering virtual objects before analyzing our experience of Being-in-a-virtual-world-in-*the*-world. I will conclude by examining the disorienting transformation as a means to explicate the rhetorical force generated by VR technology as *veri(dis)similitude* and what this unique force reveals about the human experience of objective reality in general and our understanding of that experience. This transformation, however, must be understood not as an “illusion,” but rather as an experiential and citable reality; for, to use Burke’s words, “[t]he structural relations involved” in moving a user to a mode of Being-in-a-virtual-world as well as

producing the phenomenological and ontological shock must, therefore, be “observable realities” (*GoM* 57).

The Aim, Function, and Motive of Virtual Reality Technology

We seek to win back intact the naming force of language and words; for words and language are not just shells into which things are packed for spoken and written intercourse. In the word, in language, things first come to be and are. For this reason, too, misuse of language in mere idle talk, in slogans and phrases, destroys our genuine relation to things. (Heidegger, *Introduction* 15)

I seek to articulate, similar to Heidegger’s goal indicated in the epigraph, our genuine relation to VR technology; for, indeed, as I will discuss at the end of this paper, in its functioning to move the user to a mode of Being-in-a-virtual-world-in-*the*-world the technology, necessarily, obscures our genuine relation to and understanding of VR technology in the moment. This motive of mine, as Kenneth Burke argues, “brings us to the subject of communication, since motives are distinctly linguistic products” (*PC* 35). As such, this way of thinking through VR technology will, necessarily, lead through language. More specifically, as Burke argues elsewhere: “[t]o tell what a thing is, you place it in terms of something else. This idea of locating, or placing, is implicit in our very word for definition itself: to *define*, or *determine* a thing, is to mark its boundaries, hence to use terms that possess, implicitly at least, contextual reference” (*GoM* 24). To define a technology which attempts to replicate the experience of objective reality, the object (an artificial ‘world’ which is experientially “real”) must be defined in terms of something else: the *virtual*. “An epithet,” Burke claims, “assigns substance doubly, for in stating the character of the object it at the same time contains an implicit program of action with regard to the object, thus serving as motive” (57). In light of this we might first consider virtual reality linguistically, as its epithet alone carries profound ontological implications.

Philosophically speaking, as Michael Heim observes, “virtual” is a term meaning “not actually, but as if” (220). Etymologically, “virtual” evolved from post-classical Latin’s *virtualis*: “of or relating to power or potency,” that which has the power to produce an effect (“virtual,”). More contemporarily, in reference to the technology, it is used as an epithet to qualify an artificial “reality,” and with its etymological roots and contemporary usage we are given our first inclination towards the rhetorical force generated by virtual reality technology: its motivation. VR technology and, subsequently, any virtual program is a human product and, thus, the product of a human motive. The epithet, “virtual,” should, then, both state the nature of the object (a *virtual* reality) as well as contain an “implicit *program* of action” thereby functioning as the motive of VR technology. The epithet, therefore, suggests that the *motive* of VR technology is to create an artificial ‘world’ the experience of which is so “real” that it is indistinguishable from our directly lived experience of objective reality. Rhetorically speaking, then, this *motive* implies that the *aim* of the technology is to manifest a ‘world’ which functions to move or persuade the user to accept the virtual world and their experiences within it as objectively real: to move the user to a mode of Being-in-a-virtual-world-in-*the*-world. Indeed, the *aim* is evident not only in the terms which define it, but also in the technology’s *technical* and *rhetorical function*.

What needs to be acknowledged regarding VR technology’s *technical function* is that it serves to replace or supplant virtual reality over and against that of objective reality. In order to function successfully, by its very nature, it must distract the user from objective reality, it must get the user to become more focused on the virtual simulation than their physical existence within the real world; and, the sensory deprivation is one aspect which works toward this end. For, if the real world remains dominant within or continually intrudes upon the user’s psyche, then they will never become “concernfully absorbed” to the extent necessary for the technology

to establish a virtual world over and against that of the real. Being-in-a-virtual-world-in-*the*-world is not simply a state produced by the reception of sense-data; rather, it “acts directly on our sensibility” and, as Chaim Perelman and L. Olbrechts-Tyteca argue in regard to presence, “is a psychological datum operative already at the level of perception” (116). Here I must emphasize that to “concernfully absorb” the user and move them to Being-in-a-virtual-world-in-*the*-world is to catalyze—ideally, at least—a mode of perception that fails to acknowledge the existence of the technological apparatus which generates and mediates the virtual experience. Seen from this perspective, then, for a VE to move the user to a mode of Being-in-a-virtual-world-in-*the*-world through “concernful absorption” it must construct a strong “reality ethos.” In other words, the user should feel like they are an active participant within a digitally generated, simulacral environment that recognizes and responds to their actions, the experience of which is perceived “as if it is only mediated by human senses and perceptual processes” (ISPR par. 2). What needs to be emphasized here is that a virtual environment is, first and foremost, an *environment*. Accordingly, the degree to which a user is successfully “thrown” into a VE and to which that VE “concernfully absorbs” the user depends on how it constructs and represents itself as an ecology: how it manifests the relations of the virtual objects and creatures to one another and their environment, and to the user as well. So then, if the *motive* of VR is to create an artificial ‘world’ as “real” as objective reality, then it *aims* to move the user to a mode of Being-in-a-virtual-world-in-*the*-world by constructing a persuasive ethos of reality which can only be achieved through VR’s *technical* and *rhetorical function*.

The experience of VR, at least at its current stage of development, cannot escape the apparatuses which generate and mediate it: the user must still strap an awkward HMD to their face, manipulate handheld controllers, and deal with a fifteen-foot-long cable which tethers the

HMD to the computer that generates the experience.² This is, perhaps, one of if not the most detrimental factors towards achieving a mode of Being-in-*a-virtual-world-in-the-world*; for every time the user stumbles over the cable, becomes aware of the slightly blurred resolution (commonly known as the “screen door effect”), or has to adjust the position of the HMD disrupts their “concernful absorption,” making the user aware of the necessary technological devices. Despite this, evinced in the technology’s ability to evoke instinctual physical responses (e.g. instinctively ducking to avoid a virtual object coming directly at the user), virtual worlds can be created in such a way, maximizing vivid detail and interactivity, that the user becomes so “concernfully absorbed” that they forget, if only momentarily, the technological apparatuses and the fact that the experience is nothing more than a symbolic creation: the user’s interpretation of the visual and auditory sense-data created by the programmer(s).

No user, no matter how experienced, is already in a mode of Being-in-*a-virtual-world-in-the-world* when they first dive into a VE; rather, it is a mode of perception that is rhetorically achieved over a period of time through many different factors. Yet, conversely, no one is “thrown” into a virtual world and *sees* the reality: a visual, virtual artifact is, at base, bytes of data which are rendered into pixels which compose three-dimensional (3-D) images mediated through the two displays of the HMD. The question is what do we *see* and what do we see in VR? Virtual leaves, grass, trees, we instantly identify and *see* them as such. So then, what we see (pixels mediated through lenses) is a far cry away from what we *see* (leaves, trees, and grass). Thus, the ability of VR technology to “throw” and “concernfully absorb” a user—first and foremost—demonstrates the technology’s capacity to change the user’s mode of perception; it

² Although, very recently, several devices have been commercially released which allow one to get rid of the cable and use the HMD wirelessly.

must move them from *seeing* the simulacral world *as* an artificial construct from which the user is separated to some extent to a mode of perception wherein the user perceives their physical and psychological presence as actively implicated within the virtual world as a participant whom acts and is acted on. In other words, it must replicate or mimic our basic state of “Being-in-the-world” by constructing a persuasive ethos of reality through its *technical* and *rhetorical function*. What the ability to evoke a change in the user’s mode of perception highlights, therefore, is the sheer rhetorical force exerted by a technology which is able to induce the user in a mode of Being-in-a-virtual-world-in-*the*-world, enabling them to psychologically receive an experience within the virtual world as “real” in the moment it is experienced.

I use the term “experience” specifically; phenomenologically, an HMD renders artifacts directly. We see and identify with the virtual artifacts in the graphical representation which refers us to things; however, those things which we perceive have no physical corollary in the objective space within which we experience the VE. Despite having no physical referent within the user’s objective space, phenomenologically everything experienced within a VE is received as a first-person experience. Furthermore, once the user has been “thrown” into a VE our “circumspective dealings” are illusorily perceived as the product of the user’s labor—the direct result of their conscious, active practice within the VE. Each action undertaken, consciously or not, works to psychologically harmonize the user with the VE and entities that exist within it. This harmonization or “concernful absorption,” which takes place over a period of time, then, works to move the user to a mode of Being-in-a-virtual-world-in-*the*-world. This VE now holds the potential to produce affects or changes within the user themselves.

From this perspective, a VE is not merely a neutral, isolated platform, but has the potential to *rhetorically function* as an actant. An actant denotes both human and nonhuman

actors; it, as Bruno Latour states, “can literally be anything provided it is granted to be the source of an action” (373). The focus of the concept, then, is on the relations of the various actors within a network, the associations of actants. Drawing from Latour’s theory of actants and actor-network theory, Laurie E. Gries offers the concept of “[r]hetorical actancy” as a supplement to rhetorical agency because of its emphasis on “the relations and activities,” the associations of various entities (74). Reality, from this perspective, is not constituted by any individual human or nonhuman actor; instead, it is the perception, the resulting interpretation of “human, nonhuman, material, conceptual, semiotic, and technological entities” existing within a dynamic relationship (74-5). Thus, a computer-generated virtual reality (the environment, objects and entities within it) has the ability to interact with the user in the present moment, in real-time, to use human language and symbols, and, perhaps even, can serve as the simulacrum of a social role (e.g. a teacher, a villain, a wild animal, a pet, etc.).

Kenneth Burke famously defines rhetoric as symbolic action: “*the use of language as a symbolic means of inducing cooperation in beings that by nature respond to symbols*” (RoM 43). The human mind, as he argues elsewhere, conceives reality in a somewhat roundabout way: “through various *media* of symbolism” (LSA 52). Because a VE is a purely symbolic creation which has the potential to produce a profound rhetorical affect, I believe we might think of a VE’s *rhetorical function* as a rhetorical actant which operates through symbolic action, working to induce cooperation in the user through various media of symbolism: 3-D images, 360-degree spatialized sound, text, videos, etc. Due to its ability to change the user’s mode of perception and move them to a mode of Being-in-a-virtual-world-in-the-world, such a VE has the possibility of effecting a change in the user, and, perhaps, even move the user to follow or deliberately transgress social norms (e.g. gender stereotypes, socially acceptable behavior, etc.). This

possibility, however, is dependent on the virtual program's effectively constructing a persuasive reality ethos, on the program's ability to persuasively manifest the dynamic relations or networks of involvements between various objects and entities within a specific environment. Framing a VE as a rhetorical actant which operates as symbolic action directs our attention to the various symbols, the objects, entities and actors and the relationships between them and their environment which work to construct the program's ethos of reality. However, it also directs our attention to the fact that the particular virtual program is a human construct with an intentional design. While every individual program may differ in its ultimate operation, all are intended to move the user to *act* in some way. Even a static and non-interactive VE of a real or fictional place—by literally locating them there—implicitly directs their attention to the elements which construct the VE to induce them to *actively* observe the 'world' into which they have been "thrown."

An Ethos of Reality; or, A Persuasive Virtual Ecology

Gnomes & Goblins is a human created perceptual world or a virtual simulacrum—Jean Baudrillard's notion of a representation which has no reference to an original in objective reality—and this lack of a physical referent is of a two-fold nature: first, because it is not a virtual rendition of a real place or location it could be said to have no reference to an actual earthly place; and, secondly, the specific virtual tree you see in a VE, for example, has no actual physical referent in the physical space within which the user experiences the simulation. Because a VE is a computer simulation it is not bound by the physical laws of objective reality nor is it limited to earthly objects, entities, and places. So, while one of the common applications for this

technology is to produce interactive renditions of objectively real places,³ it is possible to produce a wholly fictional VE: a digitally rendered simulacrum. The goal of rendering, as Timothy Morton argues, is to “simulate reality itself: to tear to pieces the aesthetic screen that separates the perceiving subject from the object. The idea is that we obtain an *immediate* world, a directly perceived reality beyond our understanding” (35). In other words, the goal of a simulacral VE is indeed to simulate reality and present an *immediate* ‘world’ to the user; but if that ‘world’—perceived as “real” if only for a moment—is a representation with no original in objective reality, then the “reality” which is simulated and subsequently experienced is beyond simply maintaining representational, visual fidelity to an actual object or place; rather, what is simulated is a situational reality: how the human subject, when located within a particular environment, acts and interacts with and within the ‘world’ around them according to the possibilities afforded⁴ by that environment. To successfully “throw” the user into a VE and then move them to a mode of Being-in-a-virtual-world-in-*the*-world relies on more than the visual fidelity of the VE and virtual objects within it, though this is, indeed, a factor; rather, it is in how the program manifests the relations between the user and the VE: how it frames the user’s mode of encounter with the virtual world and the objects and entities which populate it. In order for an experience to be received as “real” within a VE depends on the user achieving a state of Being-in-a-virtual-world-in-*the*-world. This mode of Being, therefore, depends upon the virtual

³ For instance, the program *SteamVR Destinations* allows the community to upload real and fictional locations which the user can explore. Some of the real locations include sections of Thingvellir National Park in Iceland, the Umayyad Palace in Jordan, and the Harborough Rocks in Derbyshire, England.

⁴ I use this term according to James J. Gibson’s definition of “affordances”: “The *affordances* of the environment are what it *offers* the animal, what it *provides* or *furnishes*, either for good or ill. The verb *to afford* is found in the dictionary, but the noun *affordance* is not. I have made it up. I mean by it something that refers to both the environment and the animal in a way that no existing term does. It implies the complementarity of the animal and the environment” (119).

program constructing a persuasive ecology—through its manifestation of objects, entities, actors and the relationships between them—and, in so doing, a persuasive reality ethos.

The basic state of Dasein in general is “*Being-in-the-world*” (BT 78). Heidegger will, later on, reformulate “Being-in-the-world” in terms of “*thrownness*,” as I have already mentioned, “*projection*,” and “*fallenness*.” Dasein “is thrown in such a way that, as Being-in-the-world, it is its “there”” (174); in other words, every situation into which Dasein is thrown is perceived as a world that, in some way or another, matters. Because the world into which Dasein is thrown *matters to*, has meaning for, Dasein, Dasein “is thrown into the kind of Being which we call “projecting”” (185). “Projection” is “the existential structure” which underlies understanding (185). Dasein projects itself onto the range of possibilities for potential action, the range of possibilities which Dasein *understands* constitutes how the world makes itself intelligible for the human. Lastly, “fallenness” is Dasein’s absorption in the common practices and processes through which we understand and interpret the world around us guided by “idle talk, curiosity, and ambiguity” (220). These three features of “fallenness,” then, characterize how the ‘world’ is, in an everyday way, made intelligible to Dasein, the way in which Dasein understands its Being as its “there.” However, “fallenness,” as an “inauthentic” mode of Being, conceals a genuine understanding with the “there” of Dasein’s Being through a fascination with and absorption in the ‘world’ of Dasein’s concern. Thus, Being-in-a-virtual world-in-*the-world* should also be formulable in such terms.

In order for a virtual program to move a user to a mode of Being-in-a-virtual world-in-*the-world*, several things need to happen first: (1) the user must be, or become, acquainted with handling the equipment (the HMD and handheld controllers which allows them to engage with the virtual world) so they are able to manipulate it in a trouble-free, and almost subconscious,

way; the user must be so familiar with using the equipment that they have no conscious awareness of the equipment *as* equipment or their operation of such. Unless the user is able to forget about the necessary presence and their use of the equipment, unless the equipment-in-use is phenomenologically transparent, then the technology will never be able to evoke a mode of Being-in-a-virtual-world-in-*the*-world. (2) To move the user to Being-in-a-virtual-world-in-*the*-world VR technology must evoke a change in our mode of encountering virtual objects. The user must come to see the virtual objects as potential items of equipment which are, to use Heidegger's phrase, "ready-to-hand" as being for a certain task (97). They must then become familiar with the mechanics for interacting with this equipment and their potential usages within the VE. (3) The program must first "throw" the user into a familiar 'world' which *matters* and then, through its *technical* and *rhetorical function*, "concernfully absorb" the user into that 'world' by constructing a persuasive virtual ecology and ethos of reality. This ecology will be persuasive if it allows the user to "project" themselves onto the possibilities afforded by the program. In other words, the possible actions allowed by the program must be familiar to those of objective reality; for if the user dives into a virtual world which only allows the user to stand in a single (virtual) place, being unable to virtually "move"—especially when the user can physically move in their objective space—the lack of this possibility for action will function as a disturbance for the user: the virtual world will be out of kilter with the projective possibilities of the real world. The "felt," physical world of objective reality (the space of our physical locomotion) will be out of kilter with our position and lack of movement within the VE. When the three steps have been achieved, the user's mode of Being is transformed from "Being-in-the-world" to Being-in-a-virtual-world-in-*the*-world. Once the user has undergone this transformation in their mode of Being, they have come to a mode of perception wherein their

critical faculties have been suspended. The user, having now “fallen” into the virtual world, has no genuine understanding of the virtual world into which they have “fallen”: the user does not *see* the reality of the technologically generated and mediated experience, but has been moved to an “inauthentic” mode of Being wherein they experience an artificial world as “real.”

Turning our attention to the virtual program *Gnomes & Goblins*, if the implicit *motive* is to manifest an artificial world which is experienced as “real,” then it will *aim* to move the user to a mode of Being-in-a-virtual-world-in-*the*-world by constructing a persuasive virtual ecology into which the user is “thrown” and, thus, a persuasive ethos of reality. It will attempt to construct this ethos of reality through its *technical* and *rhetorical function*. Therefore, as *Gnomes & Goblins* attempts to construct its reality ethos by representing itself, as persuasively as possible, as an ecological environment replete with life-like, fantastical goblins, in order to analyze this virtual ecology, I will be adopting Jakob von Uexküll’s ecological methodology for its ability to diagnose and analyze the relations between the observed subject and their specific environment. Perception, according to Uexküll, is a two-pronged process: first, the subject “imparts each object a perception mark [*Merkmal*] and with the second an effect mark [*Wirkmal*]” (49). This process thereby attributes to certain qualities of the object either “carriers” of perception or effect marks, all of which “are connected with each other through the structure of the object” (49). The first task of someone who wants to research an *Umwelt*, Uexküll notes, “consists in seeking out the animal’s perception signs” (53). However, both perception and effect signs are “always spatially bound” (54), and so, subsequently, one must also consider the particular nature of “environment spaces.” Uexküll posits that there are three types of environment space, “effect space,” “tactile space,” and “visual space” (53-63). Effect space is “the free space of our movements,” measured out by what Uexküll calls “direction steps,” which

is made aware to us “through the sensation of direction or *directional sign*” (54). Tactile space is not, like the directional step, a space of movement but one of stability, “a fixed, one place [*der Ort*]” the configuration of which is dependent upon “a perception sign belonging to the subject” and not with the nature of the subject’s surroundings (60). Visual space, lastly, is the entire space one can take in with the physical capability of sight; in this process of seeing, to each element of sight “there corresponds a place in the environment, for it so happens that one local sign is assigned to each visual element” (61).

In order to analyze how the user is “concernful absorbed” into the *Gnomes & Goblins* world into which they have been “thrown” and subsequently moved to a mode of Being-in-a-virtual-world-in-*the*-world, the structural relations for analysis derive from the artifact itself and the medium through which it is experienced. I began by examining the simulation, firstly, for perception signs within my visual space; secondly, via directional steps, the confines of my effect space; and, lastly, for effect signs and their possible functional tones. While I categorize these three aspects separately, it must be mentioned that they are dynamically interconnected, and it is, indeed, the interplay and overlapping of these features that generate the driving ecological force behind, firstly, “thrownness” and, then, “concernful absorption.” Uexküll’s methodology is beneficial for examining a VE and its virtual entities, as well as being effective in identifying the various virtual elements in their relation to the user; however, because it is geared towards observing the relations between the animal subject and its particular habitat, it is not as productive in helping us consider the human’s experience of such a virtual world nor the potential phenomenological and ontological implications such an experience might produce. If VR can phenomenologically enchant the user, moving them to interpret the illusory symbolic world as objectively real then it seems reasonable to suggest that it may also serve to disenchant

or defamiliarize the processes by which we phenomenologically interpret objective reality and understand our ontological position within it. Either possibility, though, is predicated on the technology evoking a change in the user's mode of "Being-in-the-world" of objective reality.

Such tales of miniaturization, like the transformative event experienced within *Gnomes & Goblins*, are not unfamiliar, having circulated within our entertainment culture for some time.⁵ Strands of grass as tall as trees, and acorns larger than beach balls convey enough visual information to alert the user that they are small in comparison to the size of the other objects in their environment; however, simply being placed in such an environment is not enough to produce the phenomenological disorientation or ontological disruption because it is culturally and conceptually viable, and, secondly, because the environment in which the user is placed—overlarge though it may be—is stable and remains functionally and mechanically the same.

Conversely, in *Gnomes & Goblins* the user visually experiences a change in scale but only after, and this is the key feature of the event, the user has become acquainted with their position within the VE and naturalized the processes and mechanics of interacting within it; only after the user has come to *be-in-the-virtual-world* into which they are "thrown" can the user experience the profound defamiliarization that accompanies the change in scale. The new, shall we say, goblin-sized perspective then allows the user to experience, technically, the same environment, but the meaning generated from experiencing that environment has been fundamentally altered because of the change in scale and the fact that the user has already experienced it: the status of the user shifts from something like an intrusive presence whose gaze is directed *at* the goblins and their home to someone who is invited *into* their home and

⁵ For instance, Joe Johnston's movie *Honey I Shrunk the Kids* or Lewis Carroll's *Alice's Adventures in Wonderland*, where Alice's size shrinks or grows depending on what she ingests.

encouraged to experience it from their perspective; the first position is one of exteriority, while the second is one of interiority. Though the relatively advanced state of the goblins' civilization is observable in the degree of their technological development (e.g. the existence of a library, blacksmith shop, and alcohol still), the shift in perspective also encourages a shift in perception. Because of the drastic difference in size between the user and the goblin's world, one first views the society and its level of development as something akin to a curio; however, once the user is transformed the VE becomes far more intimate and inviting. In other words, the ontological orientation or mode of Being of the user is—prompted by the phenomenological transformation and shock—transformed because our Being-in-a-virtual-world-in-*the*-world has been fundamentally altered; and, it is in this moment of disruption that the rhetorical force generated by VR technology is unconcealed.

The *function* of VR technology can be seen to consist of, as I have tried to indicate, two types: a *technical function* and a *rhetorical function*. VR functions *technically* by “throwing” the user into an artificial, symbolic ‘world’ by, firstly, sensory deprivation: depriving the user of the visual and auditory sense-data which constitute objective reality; and, secondly, by supplanting the visual and auditory sense-data of objective reality with that of the virtual world. VR functions *rhetorically* as a rhetorical actant which operates as symbolic action. The *aim* of VR technology, then, is to “throw” the user into a virtual world and, by conditioning their “circumspective dealings,” “concernfully absorb” the user into the ‘world’ and thereby move them to a mode of Being-in-a-virtual-world-in-*the*-world through its *technical* and *rhetorical function* by constructing a persuasive virtual ecology and ethos of reality. In light of this, we might now claim that the *motive* of VR technology is to create an artificial ‘world’ which is as immediately, intensely, and directly “lived” as that of objective reality, a virtual reality which can mimic our

“Being-in-the-world”—our “thrownness,” “projection,” and “fallenness”—in objective reality. Indeed, it is only because of the technology’s ability to mimic such that the disorientation felt from the transformation in *Gnomes & Goblins* can be experienced—and, subsequently, articulated—at all. However, these three elements of VR technology do not explain the source for the phenomenological disorientation or ontological disruption. Rather, as I will argue, it results from the rhetorical force generated by VR technology: its *veri(dis)similitude*.

While we have discussed the *function*, *aim*, and *motive* of VR technology at large, in order to articulate the rhetorical force generated by VR and how it affects the user we must see how these elements manifest in a specific VE. I just mentioned the *technical* and *rhetorical function* of VR; however, this needs further clarification. For the virtual world into which the user is “thrown” is, at base, a program designed and built by humans. The VR program the user experiences, then, as a completed product, is essentially static and stable. What I mean is that all of the possibilities of action for the user within a virtual world are *always already* predetermined by the programmer(s) and the program. A specific virtual program will, therefore, *technically* allow only a specific number of actions and interactions to be made with a specific number of virtual objects and entities. As every virtual program has, at least implicitly, an ultimate operative goal or design for the user, that ‘world’ is constructed in such a way as to *technically* and *rhetorically function* to condition the “circumspective dealings” of the user so that they, oftentimes unconsciously, work to fulfill the program’s ultimate design. The task of the researcher analyzing such a virtual ecology, therefore, is to make manifest the intentionally designed relations between the user, the virtual entities that are encountered, and the VE which governs the experience of the user.

A user enters a VE fully aware that what is experienced is a virtual simulation or game, fully aware that it is not “real,” so to speak. Despite this conscious understanding, well designed virtual simulations have the ability to move the user to a mode of Being-in-*a-virtual-world-in-the-world* through “concernful absorption”—and, indeed, it is the ideal for which such technologies strive. This is, potentially, problematic as it suggests that such technologies can create—or, at least, make manifest—new relations between environment, technology, and the humans understanding and experience of such. In order to account for the phenomenological experience of VR—how the human subject interprets or makes sense of such virtual artifacts—and the potential implications such an experience may hold, I will, therefore, be supplementing Uexküll’s theory and methodology with the philosophy and phenomenology of Martin Heidegger. My reasoning is two-fold: firstly, Heidegger was very much influenced by Uexküll, one of the most oft-quoted scientists in Heidegger’s writing; and, secondly, he adapted Uexküll’s theory and work to his own ends, expanding upon it in his theoretical analytic for thinking the human subject’s or Dasein’s experience with and understanding of their “Being-in-the-world” and mode of encounter with entities within that world. Therefore, taking my own experiences within *Gnomes & Goblins* as a case study, I hope to, firstly, analyze how this VE functions as a rhetorical actant designed to “throw” the user into a ‘world’ which conditions their “circumspective dealings”—through symbolic action—and “concernfully absorbs” them into the VE, all of which works to move them to a mode of Being-in-*a-virtual-world-in-the-world* by constructing an ethos of reality through its particular manifestation of the objects and entities within the VE; secondly, demonstrate the potential of VR to recalibrate the human phenomenological and ontological orientation by evoking a change in the user’s mode of perception through a reconfiguration of representations of and our experiences with reality; and,

lastly, discover the cause of the disorienting effect produced from the transformation in *Gnomes & Goblins*, the subsequent ontological implications it carries with it, and, therefore, define those implications in terms of how the rhetorical force generated by VR technology unconceals aspects of our human existence within and experience of objective reality in general.

Gnomes & Goblins: A Virtual Ecology or Umwelt

Jakob von Uexküll, an Estonian theoretical biologist for the University of Hamburg, would—over the course of his career—develop and propose a radical new perspective for zoological studies: a new way to perceive the relationship between the organism and its environment. Though he had been developing his theories since the early 1900s, they would culminate in his most famous work, *A Foray into the Worlds of Animals and Humans*, published in 1934. Uexküll was dissatisfied with the traditional scientific perspective which saw a single world within which all living organisms are hierarchically situated. Rather, he argued that every living creature exists within a species-specific, perhaps even within an individual-specific, perceptual world: “for everything a subject perceives belongs to its *perception world* [*Merkwelt*], and everything it produces, to its *effect world* [*Wirkwelt*]” (42). The receptor and effect systems constitute the organism’s “functional circle” (49)—contemporarily rephrased as a “feedback loop”—and “these two worlds, of perception and production of effects, form one closed unit, the *environment*” (42). Accordingly, there are an infinite variety of perceptual worlds as each individual’s subjective environment [*Umwelt*]⁶, metaphorically figured as a “soap bubble” which encloses the subject on all sides, is determined by their respective capacities for perception and

⁶ Though *Umwelt* literally translates as “environment,” it is closer to his thinking to define it as “surrounding world” or the “world which *environs*” a perceiving subject. This seems to be developed in response to his dissatisfaction with the concept of *Milieu*, where the environment is an “objective set of determining conditions” (Heusden 792).

production of effects. This biosemiotic⁷ perspective not only helps to blur the hierarchical distinction between animals and humans, in that all organisms are now understood to perceive and react to sense-data as signs, but also demonstrates a powerful transformation in theorizing perception. As I have previously mentioned, for a virtual simulation to *technically function* demands that the user dons the HMD and effectively cut off their sense of sight and sound from objective reality. I believe that it is useful to think about an HMD as functioning similar to Uexküll's soap bubble which encapsulates the individual user. The HMD, therefore, encompasses the user on all sides, closing "off their visual space and in which everything visible for the subject is also enclosed" (Uexküll 69). Thus, I believe it is useful to consider a VE as a virtual *Umwelt*.

HMDs use motion sensors to trace the movements of the user, simultaneously adjusting the viewing screen accordingly. These sophisticated sensors measure what Michael Heim refers to as the "'six degrees of freedom,' that is, the three dimensions (up-down, back-forward, side-to-side) of the human body, plus the angles at which the body can slant (like the pitch, yaw, and roll of air planes)" (21). In other words, the six degrees of freedom constitute the user's effect space. The HTC Vive, which I used to conduct my research, allows for what is called "room-scale" VR. By using two tracking units positioned on opposite corners of a square or rectangular space, the Vive allows the user to navigate a virtual space physically, with directional steps, rather than using a controller or keyboard. This, then, allows the user to explore and experience simulated environments as their head turnings and physical locomotion alter viewpoint and perspective. The limits of the room-scale space, a rectangular area with sides up to fifteen feet long, constitute the confines of the tactile space within which the user experiences the virtual

⁷ I must note that this term used to describe Uexküll's theory has been retrospectively applied.

simulation. When combined with controllers or data gloves, these systems allow the user to pick up and manipulate the objects one comes across in VEs. These objects, including those that are not interactive, all exist within the user's visual space, each entity being assigned a local sign. Furthermore, each object or stimulus is then either neglected or attributed a particular "functional tone": the use the user needs "to make of certain stimuli" (Uexküll 5). When a certain stimulus appears in the user's visual space it is immediately charged with a particular tone (a chair assumes a "sitting tone," for example). Thus, the combination of perception and effect signs and functional tones work to constitute the virtual *Umwelt*. As a rhetorical actant, then, this virtual *Umwelt* is comprised of the objects (inanimate things such as leaves, trees, rocks, etc.) and entities (the goblins and the user themselves) and the structural relationships between them and the VE which work to rhetorically move the user to a mode of Being-in-a-virtual-world-in-the-world.

The ultimate concluding operation of *Gnomes & Goblins* is to get the user to receive and, subsequently, ring the bell which initializes the user's miniaturization; accordingly, the *technical* and *rhetorical function* of the program, the VE, objects, entities, and types of actions and interactions which are afforded all work to condition the user's "circumspective dealings" so that they fulfill this operation. The end result of *Gnomes & Goblins* is dependent upon the user building a relationship with the shy little goblins who populate this virtual habitat. However, how the user chooses to interact with the goblin dictates how the narrative progresses. Though the user must accomplish several tasks to trigger the next narrative sequence (giving the goblin the specific fruit which triggers the goblin's gifting the bell to the user), these tasks are left to the user to discover. This is, indeed, one of the defining features of this simulation: its lack of any kind of textual or verbal cues which instruct the user how to interact with the 'world' and inform

them of what they must accomplish. Such a lack thereby directs the attention of the user to the ‘world’ itself, conditioning their “circumspective dealings” so that they explore it and attempt to interact with various objects as they so choose. In so doing, they become more acquainted with how to move and interact within the ‘world’ via the handheld controllers, slowly interiorizing the processes the goal of which is to make such interactive and operative processes second nature. This exploratory process, thus, works simultaneously as a kind of kinesthetic learning, encouraging the user to become familiar with the controllers as well as the ‘world’s’ mechanics: the interactive processes allowed by the program. As *Being-in-a-virtual-world-in-the-world* cannot be achieved until the user becomes so familiar with the hardware, virtual objects, and interactive processes that they no longer need to consciously think about them, that the equipment and processes are phenomenologically transparent, this exploratory process—our “circumspective dealings” with and within a ‘world’ which *matters to us*—and kinesthetic learning is the first conditioning process employed.

The user begins in the opening menu—an open, grassy expanse which extends into the horizon—confronted by several perception signs: a signpost that has an unlit candle appended to each direction which details a different aspect of the program. This is the only



Figure 1, *Gnomes & Goblins* Opening Menu. Wevr and Reality One, 2016. Author's screenshot.

part of the program that contains any instructions; if the user looks at their “hands” within this start menu, a textual annotation can be seen which instructs the user to pick up the single lit candle which rests on a stump in front of them and use it to light any of the unlit candles which

adorn the signpost. However, this process is more difficult than one might imagine; it takes some experimenting with manipulating the candle within their effect space before the user can, usually, successfully light the desired candle and move into the game proper. Whereas the signpost's initially conceived function tone could be something like a

“direction tone,” in that, not unlike a trail sign, it indicates two different directions, it now assumes in “interface tone.” The candle assumes a similar “interface tone”: it is transformed into an object which the user must employ to initiate the start of the virtual experience.

Lighting the candle on the direction labeled “Goblins” moves the user to the main VE they will get to explore. Transported to a fairy-like forest complete with tiny homes in hollowed-out trees and rope suspension bridges connecting the various dwellings, the user is set in this place at either day or night (the time of day alternates each time the program is run). Fireflies buzz around as the user is tempted to investigate the interior of the tiny treehouses, inviting them to push open the wooden shutters and peek inside. The user can even stick their head down a hollow stump replete with an archetypical medieval bar or pub stuffed with casks, tables and mugs. Since physical locomotion in this program is 1:1—a single step forward is just that, a single step determined by the user's stride—the explorable effect space is rather limited;

however, the miniature village, detailed foliage which sways with the wind, non-interactive objects (the rocks and roots at the user's feet, the trees in the visual space outside of the user's effect space) and interactive objects like acorns and fruit (potentially a peach or apricot) which the



Figure 2, *Gnomes & Goblins* Opening Menu. Wevr and Reality One, 2016. Author's recording.

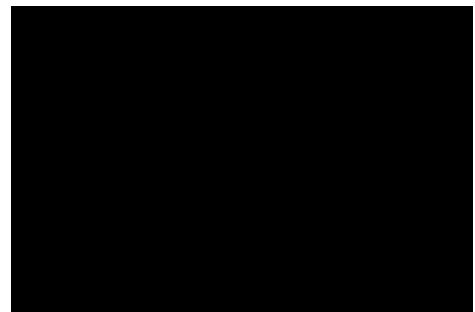


Figure 3, *Gnomes & Goblins* VE. Wevr and Reality One, 2016. Author's recording.

user can pick up help to dispel any dissatisfaction at the limited size of the VE; rather, it works to draw the user in deeper, encouraging their “circumspective dealings” to explore and experiment with what is possible within the effect space manifested in the program.

The illusive and shy goblin makes a brief appearance after the user has explored the space for several minutes; yet these appearances always seem to be on the periphery, and it is not until later that the user can get a clear look at the little creature. Several of these “cameos” take place before the goblin turns its attention to a piece of fruit hanging in one of the higher tree branches. Grunting with effort, the goblin is unable to reach the fruit and briefly goes inside to sharpen his axe before returning to cut it from the branch. Once the fruit has dropped to the ground the goblin waits for the user to pick it up and hand it over. No words are exchanged. The goblin merely looks at the fruit and then the user, and it is up to the user to figure out what the goblin wants. Upon doing so, the goblin gifts the user with a bell which shrinks them down to the goblin’s size, thereby allowing the user to now navigate the interior of the goblin’s dwelling. The preview lasts as long as the user maintains their hold on the bell, fading to black and ending once it is dropped.

At the level of perception signs, we have all those visual elements, previously discussed, one finds in the enchanted forest such as the goblin, fire flies, and hollow treehouses. The lifelike sensory details of the VE work to “throw” the user by signaling believable experiences encountered in physical reality, including the aesthetically beautiful visual representation of the world and objects as well as the 360-degree spatialized sound (an attribute commonly used to direct the user’s attention to a particular area of the VE). Though the setting is one of fantasy, the actions and events that occur within the effect space of this fantastical setting, other than the miniaturization, are common to human experience in general, and, thus, it invites the user to see,

hear, and engage the ‘world’ in a way not dissimilar from interactions in physical reality. This relationship, this commonality between acting in the effect space of reality and that of virtual reality, helps to familiarize the user to the mechanics of operation within the virtual world more quickly. Moreover, as the user’s size and spatial perspective in the program is the same as in reality (determined by their height and the HMD), the miniaturized world helps bring a sense of security, further encouraging the user to search and discover.

In a sense, then, the old adage “less is more” is at work in this program’s design. By limiting the total area of the user’s effect space to the bounds of their physically explorable space in objective reality,⁸ encourages the user to explore this small space in greater detail rather than a large space or several spaces superficially. Moreover, by limiting the total number of interactive objects (shutters, doors, acorns, apricots, sticks, and the leaves, smaller branches and bridges which sway if the user “hits” them with the controller), it directs the user’s “circumspective dealings” to various objects and entities within the virtual world and encourages them to experiment and discover which of them can be manipulated and which function solely as perception images or visual objects. This *technical* limitation of effect space and interactive objects has two primary functions: firstly, this limiting the total number of possibilities for action available to our “circumspective dealings” works to “concernfully absorb” the user by encouraging a detailed examination of the effect space and virtual objects, allowing us to become more familiar with both the objects and the mechanics of the ‘world.’ Secondly, limiting both the total effect space and number of interactive objects, the user more quickly discovers the limitations of the VE—what is or is not afforded to the user by the virtual program—as there are

⁸ When you set up the HTC Vive tracking units to allow for room-scale VR, you can create a rectangular space anywhere from 1.5x1.5 meters to 5x5 meters. *Gnomes & Goblins* requires a space of at least 2x1.5 meters. Accordingly, the effect space of the VE matches the size of the user’s room-scale settings.

only a small number of possible interactions with an equally small number of objects and entities within an effect space of limited size; furthermore, this allows the user to become familiarized more quickly and accomplish the necessary tasks which lead to the culminating transformation more smoothly. For instance, it is likely that the user has picked up one of the acorns and gifted it to the little goblin long before the creature chops down the key fruit which, when given to the goblin, constitutes the culminating act for the user. Thus, when the key fruit is chopped down, the user is predisposed to give it to the creature.

With no textual perception signs of any kind and the only auditory signs being those of the “natural” environment—the wind blowing through the leaves, the crackling brush and foliage as the user navigates, or the squeak of a hinge as a shutter is pushed open—the completion of *Gnomes & Goblins* is entirely dependent on the user. While there are certain clues provided, the completion of the task needed to trigger the next narrative sequence (giving the goblin the fruit and ringing the bell) is left to the user to discover, and the discovery of both task and solution can only be achieved after the user has begun to explore the effect space of the VE. When the user accepts the invitation and participates in the virtual environment, they are more likely to become “concernfully absorbed” and direct their attention away from the hardware and their actual existence in objective reality because everything is received as first-person experience; in other words, the user must first make the decision—conscious or otherwise—to approach and pick up the key fruit and *then* physically walk over to and use the handheld controllers to do so; they must also interpret the possibility of giving the goblin the fruit—a possibility which should have been discovered in the user’s “circumspective dealings” with the VE—before physically giving it to the little creature. Such a design is intended to let the user become more accustomed with moving and interacting in the virtual *Umwelt*, while the, illusory, freedom of choice

amplifies their “concernful absorption.” This “less is more design” works to condition the user’s “circumspective dealings” so that they complete the key act in a subconscious way, in a mode of perception that fails to realize that their actions are being conditioned by the program to, as subtly as possible, make sure they eventually fulfill the program’s ultimate operation. For if this culminating act the user must fulfill needs to be explained or addressed for the user by any external intrusion into the VE (a textual or auditory instruction, for instance), this would disrupt the ethos of reality the VE has theretofore been working to construct.

“Anything and everything that comes under the spell of an environment, Uexküll asserts, “is either redirected and re-formed until it becomes a useful carrier of meaning or it is completely neglected (144). Accordingly, if this program intends to “throw” the user into a VE that *matters to* them and then “concernfully absorb” the user by representing itself, persuasively as possible, as a “real” ecology, then the user, perceived by the entities within the virtual *Umwelt*, must become a carrier of meaning for those entities. At first, the goblin “perceives” you from afar or hidden behind windows and high grasses. It will eventually venture closer to the user, but only on the condition that the user remains still. Any sudden movements will send it scampering away in fright. This suggests that the user is an entity and/or agent in the virtual *Umwelt* for the goblin just as the goblin is for the user, both carrying perception and effect marks as well as several different functional tones. Just as how the meaning of the sea anemone for the crab, in Uexküll’s example, changes depending on the “mood” of the crab (the anemone becomes a “protective tone” for the crab if under attack by a squid or assumes a “feeding tone” if the crab is starving) (93), so too does the user take on different tones depending on the “mood” of the goblin. Thus, if the user abruptly approaches the little goblin, they assume a threatening meaning and takes on something like an “escape tone.” If the user, on the other hand, approaches the goblin slowly,

kneeling down to appear smaller in size, and offers the little creature a piece of fruit or an acorn, the goblin will, every so cautiously, approach the user and accept the offering. In this instance, the meaning of the user is transformed and assumes a “helpful tone.” Therefore, the “perception image provided by the sensory organs,” Uexküll posits, can “be completed and altered by an ‘effect image,’ which is dependent on the next action that takes place” (93). The user becomes a carrier of meaning for the goblin as soon as they enter into the goblin’s visual space, and, similar to the transformation of meaning the candle undergoes when the user realizes its true “interface” functional tone at the start of the simulation, the user’s functional tone for the goblin can change depending on the user’s actions.

To create a persuasive ethos of reality the program, then, seems to position, at least superficially, both the user and goblin in agential roles, in a position wherein their “actions” affect or produce a reaction from the entities within the VE. Thus, when we act and give the creature the key fruit, the creature “acts” and gifts us the magic bell. The goblin’s “actions,” though, are nothing more than compulsive motion, nothing but programmed responses triggered when the user does something the program recognizes. Despite the fact that we *know* that the goblin is a fictional creature, the goblin’s “actions” are designed in such a way that they are recognizably *familiar*; by mimicking the attributes and responses familiar to our real experiences with similarly timid animals, the program directs the user’s attention toward developing the relationship between them and the goblin and away from the fact that any of the goblin’s “actions” or “reactions” are *always already* predetermined. Even though the goblin appears to “act” autonomously, it is only as a superficial agent. The *motive* behind the goblin’s “actions,” therefore, is not its own, but rather the programmer(s). By “throwing” the user into such a virtual ecology the goblin’s “actions,” consequently, function to further condition the user’s

“circumspective dealings” so that they complete the program’s ultimate operation while simultaneously, through those conditioned “circumspective dealings,” work to “concernfully absorb” the user into a mode of *Being-in-a-virtual-world-in-the-world* wherein they see themselves and the goblin as both equally active and implicated in the VE.

Once the user has assumed a “helping tone,” the goblin rewards them with a gold filigreed bell. A common everyday object that most every user will have interacted with before, the user instinctually ascribes it a “ringing tone.” Upon ringing it, the fireflies awaken in great swaths of varicolored light which swirl around the user, obscuring their vision. During this inundation of visual and auditory stimuli the user is dramatically miniaturized down to the goblin’s size. Once this event has taken place, the bell’s

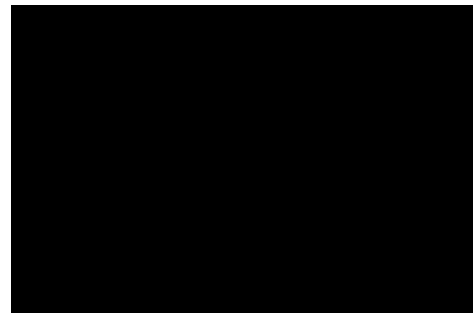


Figure 4, *Gnomes & Goblins* Miniaturization. Wevr and Reality One, 2016. Author's recording.

original operative meaning as a noise producer has been fundamentally altered; now it assumes a new meaning expressed as a new characteristic: a “magical tone.” After this transformation, any time the user rings the bell they are transported to various locations inside the goblin’s dwelling,



Figure 5, *Gnomes & Goblins* Post-Transformation. Wevr and Reality One, 2016. Author's screenshot.



Figure 6, *Gnomes & Goblins* Post-Transformation. Wevr and Reality One, 2016. Author's screenshot.



Figure 7, *Gnomes & Goblins* Post-Transformation. Wevr and Reality One, 2016. Author's screenshot.

enabling them to explore both the original and now a new effect space but from the size and perspective of the goblin. The meaning of the bell changes once again, assuming, now, a “teleportation tone.”

Though the transformative experience is disorienting, it is meant to be. By the time the transformation happens, the user should have grown acquainted with the perception and effect marks which constitute the virtual *Umwelt*. Despite the sudden and unnerving sensation, the miniaturization produces, this phenomenological disorientation does not shatter the user’s “thrownness” in the VE with its reconfiguration of the representational *Umwelt*’s perception signs, but amplifies it since everything else—the mechanics, entities, and interactive objects—remains functionally the same. The ethos of reality constructed by the ecological relations between the user and the virtual *Umwelt* allows such a magical transformation to be perceived as within the realm of possibility permitted by the fantastic theme of the setting; and it is, moreover, the ecological relations that are established which constitute the most interesting element of the goblin: for when the user is perceived or when they pick up an object the goblin sees it and it has meaning for the creature. Is it a threat or a gift? If you step forward, it moves back; if you move back it will step forward, that is, if it perceives you as a “helping tone.” It is the user’s involvement within the virtual *Umwelt* that effects the semiotic sequence; and a successful interaction with an aspect of the *Umwelt* furthers the effectiveness of the functional cycle and ecological relationship between the user and the VE. Thus, to “throw” the user into a virtual world and, by conditioning their “circumspective dealings,” “concernfully absorb” them into a mode of Being-in-a-virtual-world-in-the-world, the ontological status of the virtual environment and the objects within it need not refer to any factual, objective reality; rather, such a mode depends upon the program constructing an ethos of reality: upon, firstly, being “thrown” into a

‘world’ that has meaning for them and then the user’s “concernful absorption” into a persuasive situational reality, an effective ecological relationship with an environment which appears to act as much on the user as the user acts on it. Or, to phrase this differently, the user’s Being-in-a-virtual-world-in-*the*-world depends upon their “thrownness” into a ‘world’ which *matters*—a ‘world’ which is culturally and mechanically *familiar*—and implicates the user in a persuasive functional circle within the virtual ecology.

Our Mode of Encountering Virtual Objects and Entities

If a virtual program *aims* to move the user to a mode of Being-in-a-virtual-world-in-*the*-world it must first evoke a change in our mode of encountering virtual objects and entities. To recall one of Uexküll’s examples, the meaning the sea anemone has for the crab depends upon the mood of the crab: if the crab is starving, the sea anemone takes on a “feeding tone,” whereas if the crab is threatened, the sea anemone takes on a “protection tone.” Thus, in the case of our encounter with the goblin, though Heidegger would almost certainly disagree, we are no different from the goblin or crab. Indeed, this appears to be one instance where Heidegger draws from and adapts a concept of Uexküll’s. Heidegger argues that Dasein’s Being as “there” in the world is (BT 173), primordially, disclosed to Dasein in “our mood, our Being-attuned” (172). How the virtual world opens to us (is disclosed) depends upon our having a mood, our “‘how one is’” (173). “*The mood has already disclosed, in every case,*” as Heidegger continues, “*Being-in-the-world as a whole, and makes it possible first of all to direct oneself towards something*” (176); and, as such, we are always in one mood or another, we can never be free of moods. How we understand a virtual world, then, depends on our mood, on how the virtual world is disclosed to us. If our mood “arises out of Being-in-the-world, as a way of such being” (176), then our mood which arises out of Being-in-a-virtual-world-in-*the*-world is conditioned—to some

extent—by how the program positions the user and manifests the structural relations between them and the VE: how it locates them within a persuasive virtual ecology and what projective possibilities that VE affords the user. Let's say the mood which arises from our Being-in-the-*Gnomes & Goblins*-world is one of childish delight and curiosity. This mood will disclose the virtual world in a certain tone, to borrow Uexküll's term: safe, inviting, interesting, novel, playful. The world being opened to me in such a way, I notice a curiously shaped rock lying on the ground. I reach down to pick it up only to find that I am unable to interact or manipulate the object in any way, only to find that the object is resistant or entirely unserviceable. The fact that I am affected by this resistance or unserviceability, the fact that the virtual rock can even be perceived as having such characteristics, is "ontologically possible only in so far as Being-in as such has been determined existentially beforehand in such a manner that what it encounters within-the-world can "*matter*" to it in this way" (176).

Now aware that no structural relations between the user and the virtual rock exists, that such virtual rocks function merely as perception images, I will not likely reattempt to interact with such. These rocks will now, for the most part, largely be ignored as they do not *matter to* me in the same way because the relationship between the virtual rock and myself is characterized as unserviceable if not non-existent. What this illustrates, therefore, is how a virtual object can undergo a phenomenological transformation of sorts dependent upon a shift in our mode of encountering such objects: initially perceived as, what Heidegger calls, "ready-to-hand," it is now perceived as "un-ready-to-hand." "We discover its unusability, however, not by looking at it and establishing its properties, but rather by the circumspection of the dealings in which we use it" (102). What this phenomenological transformation in the virtual rock's mode of Being suggests is that the user or Dasein's mode of Being undergoes a similar phenomenological

transformation: Dasein becomes a subject over and against a world of objects, some of which *matter* or have meaning while others afford merely perception signs. This phenomenological transformation of both the virtual rock and Dasein, however, is dependent on the disruption (the event when the virtual rock's mode of Being shifts from "ready-to-hand" to "un-ready-to-hand") which brings to Dasein an awareness of its unserviceability. This disruption makes us consciously aware of the virtual rock as having such a characteristic; and, this disruptive event and the phenomenological transformation it produces occurs not out of a detached observation of the object, but rather through our "circumspective dealings" with it.

We ordinarily encounter entities, or so Heidegger argues, as "*equipment*"; that is, we encounter entities, "in concern," as being for a specific task: "equipment for writing, sewing, working, transportation, measurement" (97). This is, indeed, the manner which constitutes our most primordial relationship with such entities, not in a detached observation of the "thing-in-itself" but in our skillful manipulation of it: "In dealings such as this, where something is put to use, our concern subordinates itself to the "in-order-to" which is constitutive for the equipment we are employing at the time" (98). What is most interesting about Heidegger's conception of "readiness-to-hand" is that, while engaged in a skillful manner with no disruptions, equipment-in-use become phenomenologically transparent: Dasein has no conscious experience of the tools as "things-in-themselves." While constructing a table in a skilled and trouble-free way, the craftsman has no conscious experience of the hammer, nails, or wood as they would were they to sit and merely contemplate those items. Furthermore, once engaged in this skillful activity, the craftsman or user themselves become absorbed in their activity. This "concernful absorption in whatever work-world lies closest to us" is such that the craftsman has no conscious

understanding of the subject-object dichotomy, no awareness of themselves as a subject contrasted with a world of “things” and objects (101).

I have already mentioned several of the interactive objects within *Gnomes & Goblins*, the window shutters, for instance. In a Heideggerian sense, these shutters, then, can be viewed as an equipment, the “ready-to-hand” character of which must be discovered through our “circumspective dealings” with the virtual world. However, there is, taken strictly for Heidegger, “no such thing as *an* equipment” (97); rather, the character of an equipment which is “ready-to-hand” is an “*involvement*” (115). “Involvement” is used to define the relation of equipment to Dasein’s everyday activities. This “involvement,” though, is not a single structure which stands alone, but exists in a network or “totality of involvements” (116). Thus, in Heidegger’s example, the “ready-to-hand” hammer is involved “in hammering”; the hammering is involved “in making something fast”; making something fast is involved “in protection against bad weather; and this protection ‘is’ for the sake of a possibility of Dasein’s Being” (116). The contexts within which equipment is used in an everyday manner is summarily constituted by such totalities of involvements and, more importantly, these totalities of involvements, therefore, define an entity as equipment. The shutters within the VE are only intelligible as such in relation to their function of opening and closing or of concealing and unconcealing what lies visible on the other side. Heidegger’s hammer is only intelligible to Dasein as a network of significances, it is only intelligible as a hammer in relation to the other items of equipment involved and the act of producing shelter. To take this a step further, then, such totalities of involvements could be understood to constitute, in one of Heidegger’s uses of the term, “worlds”: “indeed ‘world’ can become a term for any realm which encompasses a multiplicity of entities: for instance, when one talks of the ‘world’ of a mathematician, ‘world’ signifies the realm of possible objects of

mathematics” (93). Thus, it is how the totality of involvements manifest the ecological relationship between the user and the VE that determines the realm of possible objects, actions, and interactions of a specific virtual world and whether or not those features, and that ‘world,’ become intelligible to the user as a network of significances.

Initially perceived, with light leaking through the gaps between the wooden slats, these virtual shutters are assigned a perceptual mark. Immediately after, they become a carrier of an effect mark with an “open/close” or “conceal/unconceal” functional tone. What this experimental process reveals, as Uexküll states, is that we become aware of how we instinctively, and perhaps instantly, assign effect images “for all the acts we perform on objects in our environment, and that we alloy this effect image so effectively with perception images we receive from our sense organs that the two acquire a new characteristic, which announces its meaning to us” (94-5); and, how this world announces this meaning to us depends on our mood, on how the virtual world is disclosed to us. The resulting effect mark and functional tone attributed to the shutters, therefore, is not innately received; rather, it is first suggested due to our knowledge of and past experiences with shutters which is then confirmed through a process of observing and experimenting with them in the virtual *Umwelt*. In other words, through their “circumspective dealings” the user is able to confirm which virtual objects are carriers of effect marks and which are only carriers of perception marks. What Uexküll’s observation implies is that the process by which we assign the virtual shutters, first, a perceptual mark and then an effect mark with an “open/close” and “unconceal/conceal” functional tone is historically and culturally conditioned. Analyzing the VE within the concept of Uexküll’s *Umwelt* and functional cycle, therefore, reveals that both the perception marks and effect marks of virtual entities perceived by the user are, in Uexküll’s words, “expressions of the subject and that the qualities of the objects included in the functional

cycle can only be referred to as their vehicles” (125). This position of Uexküll’s seems to be consistent with Heidegger’s “for-the-sake-of-which,” one of, if not, the most important elements in any involvement. If these two positions correspond, and if the perceptual process Uexküll identifies are culturally and historically conditioned, then this suggests that the totality of involvements which defines the virtual shutters as equipment “ready-to-hand” or the virtual rock as “un-ready-to-hand” are, similarly, culturally and historically conditioned.

Involvements, as Heidegger points out, are not uniform but are structures consisting of various elements. “The “for-the-sake-of-which” signifies an “in-order-to”; this in turn, a “towards-this”; the latter, an “in-which” of letting something be involved; and that in turn, the “with-which” of an involvement” (*BT* 120). Turning this perspective on my experience of and work with virtual reality, it could be described as the following: I (“Dasein”) am currently working with a virtual reality technology (a “with-which”), in the practical context of my room (an “in-which”), in order to analyze a virtual environment (an “in-order-to”) which is directed towards phenomenologically and ontologically explaining the experience of a mode of Being-in-*a-virtual-world-in-the-world* in such an environment (a “towards-this”), for-the-sake-of completing my MA paper and fulfilling a requirement for my PhD, that is, for the sake of my being a professional scholar (a “for-the-sake-of-which”).⁹ The last involvement in this example, the “for-the-sake-of-which,” is crucial for Heidegger as it is the foundation upon which all totalities of involvement are built. “The relational character which these relationships of assigning possess, we take as one of *signifying*. In its familiarity with these relationships, Dasein ‘signifies’ to itself: in a primordial manner it gives itself both its Being and its potentiality-for-Being as something which it is to understand with regard to its Being-in-the-world” (120). The

⁹ For more on these various features analyzed by Heidegger, see *Being and Time* 114-120.

significance of this position is twofold: firstly, it suggests that Dasein's possibilities for Being, and its practical actions or activities, are inextricably intertwined with understanding and interpretation, with the way in which objects and entities become intelligible for Dasein; secondly, because every "for-the-sake-of-which" is the foundational structure of a totality of involvement, it not only functions to define a particular object as equipment but also reflects one of many potential ways for Dasein to *Be* (a professional scholar, in the case of the above example).

Equipment cannot be known outside the totality of involvements which brings that object into intelligibility for Dasein. Thus, our "Being-in-the-world," Heidegger concludes, "amounts to a non-thematic circumspective absorption in references or assignments constitutive for the readiness-to-hand of a totality of equipment. Any concern is already as it is, because of some familiarity with the world" (107). In other words, Dasein's encounter with entities and objects occurs within a world that is *always already* familiar to some extent. "In this familiarity Dasein can lose itself in what it encounters within-the-world and be fascinated with it" (107). Therefore, set in an enchanted virtual forest which is already culturally familiar and as the mechanics for interacting within this 'world' are familiar to those of reality in general and VR more specifically, while I am engaged in a skilled and trouble-free navigation of a virtual environment the HMD and controllers—and the role they play in my academic research—are transparent facets of my overall experience; that is, as equipment-in-use they are phenomenologically transparent. But if the power were to go out and the HMD shut down, if its mode of Being becomes "un-readiness-to-hand" (103), then I become aware of the HMD as an entity of agency "with-which" I was engaged, in the context of my room, "in-order-to" act and analyze a VE

directed “towards-this” purpose of explaining the experience of Being-in-a-virtual-world-in-*the*-world.

This “concernful absorption,” however, does not imply a total lack of awareness, but rather Dasein’s awareness that is present—Heidegger’s “*circumspection*” (98)—“has a function of discovering; and it is essential to this function that, depending upon the way in which we are absorbed, those entities within-the-world which are brought along [*beigebracht*] in the work and with it (that is to say, in the assignments or references which are constitutive for it) remain discoverable in varying degrees of explicitness and with a varying circumspective penetration” (101). The loss of power disrupts the totality of involvements which constitute the virtual world, the network of total significances; it thereby disrupts my “concernful absorption” in my “work-world,” dispelling the phenomenological transparency of equipment-in-use and making me aware that my actions and mode of Being are directed for-the-sake-of my being a professional scholar. Accordingly, such disruptions can reveal the totality of involvements, or ‘world,’ which organize my behavior. Or, as Heidegger phrases it, “[t]hat which has been circumspectively taken apart with regard to its “on-order-to”, and taken apart as such—that which is explicitly understood—has the structure of *something as something*”; when Dasein, then, encounters “what is environmentally ready-to-hand by interpreting it circumspectively, we ‘see’ it *as* a table, a door, a carriage, or a bridge” (189). Thus, we *see* the virtual shutters *as* shutters. Furthermore, as the “for-the-sake-of-which” structure is the base structure of any totality of involvements, and as the network of totalities of involvements constitute a ‘world,’ this posits that the world of which we are phenomenologically aware—whether it be virtual or real—is one which is, similarly, culturally and historically conditioned.

True Presence in Virtual Reality; or, Being-in-a-virtual world-in-*the*-world

That which can be Articulated in a disclosure by which we understand, we call “meaning”. The *concept* of *meaning* embraces the formal existential framework of what necessarily belongs to that which an understanding interpretation Articulates. *Meaning is the “upon-which” of a projection in terms of which something becomes intelligible as something; it gets its structure from a fore-having, a fore-sight, and a fore-conception.... Hence only Dasein can be meaningful [sinnvoll] or meaningless [sinnlos].* That is to say, its own Being and the entities disclosed with its Being can be appropriated in Understanding, or can remain relegated to non-understanding. (Heidegger, *BT* 193)

Considering that a VE is a human product, it, compared to the natural world, is far simpler and less complex than any objectively real environment. Thus, if Heidegger’s theorizing of Dasein’s mode of encounter with entities and “Being-in-the-world” holds true in objective reality, then it should also hold true in our experience of virtual reality.¹⁰ A VE does not offer undifferentiated information, but rather an already interpreted ‘world’ that offers a predetermined number of opportunities to each individual user. Though he uses it to explicate his concept of the new, one of Theodor Adorno’s examples will serve to illustrate this point: “The relation to the new is modeled on a child at the piano searching for a chord never previously heard. This chord, however, was always there; the possible combinations are limited and actually everything that can be played on it is implicitly given in the keyboard” (32). Because it is a meticulously, human programmed ecology, all of the possibilities of action for the user are *always already* contained within the experience but are dependent upon their ability to discover them through their “circumspective dealings”; thus, when entering a new VE the user is in the same position as the child at the piano. To reiterate, one facet of Being-in-a-virtual-world-in-*the*-world is achieved

¹⁰ Interestingly, a specific VE or virtual reality in general may offer itself as a useful way for interrogating such ontological and phenomenological notions or as a means of testing such hypotheses. For if it does not prove to be true of a virtual world the possibility of it being so in objective reality, due to its incalculably greater complexity, is practically impossible. Though I cannot fully explore these implications in this paper, it might be useful to use VR technologies as a means of further testing such philosophical positions.

when the user becomes so “concernfully absorbed” that their mode of perception fails to be consciously aware of the computer, HMD, and handheld controllers that generate and mediate the experience.

The success of a VR system in creating a persuasive virtual ecology, an interactive functional circle between the user’s perception and effect world within the virtual *Umwelt*, therefore, depends on the ease of decoding and interpreting the virtual representation. It depends on the ease with which the user is able to apprehend and interpret the perception and effect signs as well as their functional tones; or, to use Heidegger’s term, the totality of involvements which manifest the virtual world and, therefore, govern the user’s behavior. That is, not until the user is “thrown” into a ‘world’ that is conceptually and mechanically *familiar* can the virtual objects be perceived as “ready-to-hand.” Only when the virtual objects are perceived as “ready-to-hand” can the functional circle be established. Not until the functional circle is established can the user’s “circumspective dealings” be conditioned and thereby be “concernfully absorbed” into the ‘world’; and, not until the user is “concernfully absorbed” can they be moved to a mode of perception wherein virtual world and entities that populate it are understood as Being “ready-to-hand”: a mode of Being-in-a-virtual-world-in-*the*-world. Our “concernful absorption” subordinates the equipment to the “in-order-to,” wherein, as a structure, “there lies an *assignment* or *reference* of something to something” (BT 97). Being-in-a-virtual-world-in-*the*-world, to adapt Heidegger’s words, is a mode of perception in which Dasein or the user “explicitly addresses itself as “I here”; it is a mode of perception which “is “yonder” which is the dwelling-place of Dasein as *concern*” (155). Being-in-a-virtual-world-in-*the*-world, therefore, is an *event* which occurs when the user comes to *dwell* within the VE, when the user is “concernfully absorbed” or appropriated by the virtual world by locating (or “throwing”) them within

structures and patterns, or practices, of understanding and acting which are familiar thereby transforming the user's intelligibility of the relationship between themselves, the VE, and the hardware generating and mediating the experience.

The basic state of Dasein in general, as I mentioned earlier, is "*Being-in-the-world*" (78). Heidegger will, later on, reformulate "Being-in-the-world" in terms of "*thrownness*," "*projection*," and "*fallenness*." Dasein "is thrown in such a way that, as Being-in-the-world, it is its 'there'" (174); in other words, every situation into which Dasein is thrown is perceived as a world that, in some way or another, *matters to it*. Because the world into which Dasein is "thrown" *matters*, has meaning, Dasein, "is thrown into the kind of Being which we call 'projecting'" (185). "Projection" is "the existential structure" which underlies understanding, and, as Heidegger posits, "[u]nderstanding is the existential Being of Dasein's own potentiality-for-Being; and it is so in such a way that this Being discloses in itself what its Being is capable of" (185; 184). Understanding, as what Heidegger calls a "disclosure," thus, always "pertains to the whole basic state of 'Being-in-the-world.'" As a potentiality-for-Being, any Being-in is a potentiality-for-Being-in-the-world. Not only is the world, *qua* world, disclosed as possible significance, but when that which is within-the-world is itself freed, this entity is freed for *its own possibilities*" (184). The intelligibility of an object, for Dasein, is dictated by its totality of involvements, the base structure being the "for-the-sake-of-which." The "ready-to-hand" is, therefore, "discovered as such in its *serviceability*, its *usability*, and its *detrimentality*. The totality of involvements is revealed as the categorial whole of a *possible* interconnection of the ready-to-hand" (184). Thus, if "projection" is the basic structure of understanding, then understanding is the process by which Dasein, as "thrown" into a world that *matters to it*, "projects" itself onto the possibilities of entities within the world and also the world as the total

context of possible significance. “With equal primordiality the understanding projects Dasein’s Being both upon its “for-the-sake-of-which” and upon significance, as the worldhood of its current world” (185). Understanding—as “projection”—is, therefore, constitutive of Dasein’s “Being-in-the-world” in regard to the way the world is disclosed—the way it opens up—to Dasein, the way it makes itself intelligible.

Understanding, then, manifests itself differently in different domains. In the realm of the “ready-to-hand,” it seems to manifest mainly as the skilled and trouble-free activity of equipment-in-use; it can manifest as an assertion of language when Dasein attributes to a “present-at-hand” object a specific quality such as weight or length; or, to recall my earlier example of a power outage, it manifests as an act of interpretation when I take the HMD, because of the disruption, explicitly *as* an HMD. “To say that “circumspection discovers” means that the ‘world’ which has already been understood comes to be interpreted” (189). To recall the epigraph which introduced this section, this interpretation, therefore, in every case is grounded “in *something we have in advance—in a fore-having,*” “in *something we see in advance—in a fore-sight,*” and “in *something we grasp in advance—in a fore-conception*” (191); and, indeed, this tripartite category seems to, for Heidegger, constitute the basis of *meaning*, of all knowing, understanding, and perceiving. The transformation that takes place in *Gnomes & Goblins*, therefore, cannot be understood in the moment as it is a projective possibility of which there is no “*fore-having,*” “*fore-sight,*” or “*fore-conception*” for the user.

The third feature of Heidegger’s reformulation of “Being-in-the-world” is “*falling*” or “*fallenness*”: “Dasein has, in the first instance, fallen away [abgefallen] from itself as an authentic potentiality for Being its Self, and has fallen into the ‘world’. “Fallenness” into the ‘world’ means an absorption in Being-with-one-another, in so far as the latter is guided by idle

talk, curiosity, and ambiguity” (220). “Idle talk, curiosity and ambiguity,” therefore, “characterize the way in which, in an everyday manner, Dasein is its ‘there’—the disclosedness of Being-in-the-world” (219). “Idle talk,” reductively paraphrased, is Dasein’s conversing, either written or spoken, in a way that does not critically examine the information of what has been conveyed; it is an “understanding which has thus already been “deposited” in the way things have been expressed,” and pertains as much to the discoveredness of entities as it does to one’s understanding of Being and its possibilities (211).¹¹ “Curiosity” searches for—rather than a belonging or what Heidegger sometimes calls a “dwelling”—endless stimulation and novelty; “curiosity is concerned with the constant possibility of *distraction*. “Curiosity” has nothing to do with observing entities and marveling at them ... To be amazed to the point of not understanding is something in which it has no interest” (216).¹² Lastly, “ambiguity” is when Dasein has lost the critical sensitivity to distinguish between “what is disclosed in a genuine understanding and that which is idle chatter” (217).¹³ The connection that must be noted between these three features of “fallen” “Being-in-the-world” is that they work to conceal any genuine understanding of the world, by concealing this genuine understanding through a fascination with and absorption in the world. For in “fallenness” “Dasein has, in the first instance, fallen away [abgefallen] from itself as an authentic potentiality for Being its Self, and has fallen into the ‘world’” (*BT* 220); Dasein has “fallen” into an “inauthentic” Being.

So then, if “idle talk,” “curiosity,” and “ambiguity” characterize the disclosedness of Dasein’s everyday “Being-in-the-world,” if they characterize *how* Dasein is its “there,” then they also characterize, to some extent, Dasein’s mood. If this is indeed the case, then the ‘world’ as it

¹¹ For more on “idle talk,” see Heidegger’s *Being and Time*, 211-14.

¹² For more on “curiosity,” see Heidegger’s *Being and Time*, 214-17.

¹³ For more on “ambiguity,” see Heidegger’s *Being and Time*, 217-19.

is disclosed and made intelligible to Dasein is, essentially, a shared world; thus, as Heidegger states in his discussion of the “they”, “[i]f Dasein is familiar with itself as they-self, this means at the same time that the “they” itself prescribes that way of interpreting the world and Being-in-the-world which lies closest. Dasein is for the sake of the “they” in an everyday manner, and the “they” itself articulates the referential context of significance” (167). In other words, the “they” articulates Dasein’s interpretation of the world, of the totality of involvements, of “the referential context of significance” and, therefore, how we understand and interpret the world, how entities become intelligible to us is culturally and historically conditioned. Furthermore, as a mood is fundamentally prior to Dasein’s understanding and interpretation, if a mood “*implies a disclosive submission to the world, out of which we can encounter something that matters to us*” (177), then what we conceive of as “mattering,” how we understand the totality of involvements which constitutes our ‘world’ of significances is similarly conditioned.

As “curious,” “Dasein lets itself be carried along [mitnehemne] solely by the looks of the world”; the kind of Being which characterizes “curiosity” is one in which Dasein “concerns itself with becoming rid of itself as Being-in-the-world and rid of its Being alongside that which, in the closest everyday manner, is ready-to-hand” (216). For Dasein, as “curious,” to dive into VR is to concern itself with escaping itself as “Being-in-the-world” (of objective reality) and the totalities of involvements which manifest that world and dictate *how* Dasein is alongside that which is “ready-to-hand” in the closest everyday manner. So then, if “idle talk,” “curiosity,” and “ambiguity” characterize the everyday way in which Dasein has “fallen” into the virtual world, if they characterize “Dasein’s absorption in the “they” and its absorption in the [virtual] ‘world’ of its concern,” this suggests, then that Dasein’s “falling” into a virtual world manifests “something like a *fleeing*” or “turning-away” of Dasein “in the face of itself” as “Being-in-the-world” (229).

By turning away from “Being-in-the-world” Dasein turns towards the virtual world and entities within-the-virtual world by “concernfully absorbing” itself in them through its “circumspective dealings.” For if the user is moved to a mode of Being-in-*a-virtual-world-in-the-world*, then the technology has, fundamentally, displaced them from their “Being-in-the-world” of objective reality.

Let us now consider Being-in-*a-virtual-world-in-the-world* for the average user in relation to VR technology’s *function*, *aim*, and *motive*. The average user is one who does not enter a respective VE with the motive to critically analyze the structural relations of that environment; rather, the average user is someone who enters a particular VE for curiosity, distraction, and/or entertainment with a predisposition to direct their attention away from objective reality for the sake of enjoyment. The user discerns the concrete situation of the VE into which they dive, as Burke would argue, “by means of the particular vocabulary of the cultural group into which” they are born (*PC* 35). Or, as Heidegger argues, “[o]nly where the word for the thing has been found is the thing a thing. Only thus *is* it. Accordingly we must stress as follows: no thing *is* where the word, that is, the name is lacking. The word alone gives being to the thing” (“The Nature of Language” 62).

If, therefore, the *motive* of VR technology is to create an artificial ‘world’ the experience of which is indistinguishable from that of objective reality, then, in order to achieve its *aim*—moving the user to Being-in-*a-virtual-world-in-the-world*—that ‘world’ must be linguistically, culturally, and mechanically viable: it must construct a persuasive ethos of reality into which the user is “thrown.” The persuasiveness of the program’s reality ethos depends on the *technical* and *rhetorical function*, on how it manifests the VE’s totality of involvements or functional circle within which the user is “thrown.” While the *technical* (locating the user—through sensory

deprivation—in a human constructed virtual world) and *rhetorical* (serving as a rhetorical actant which operates as symbolic action) *function* of VR technology as a whole remains the same, how those *functions* manifest in a specific VE will differ. For each VE will only *technically* allow the user to act in a certain number of ways with a certain number of objects; in other words, the totality of involvements, the structural relations which manifest the user's 'world' or total network of possible actions and significances, constitute the *technical* and *rhetorical function* of a specific VE. How the totality of involvements manifests within a specific VE, then, not only constructs the VE's virtual ecology but also determines the persuasiveness of that ecology, the persuasiveness of its reality ethos. If a VE is to construct a persuasive reality ethos through its virtual ecology, then that ecology and the structural relations which exist between it and the user must be *familiar* to some extent: for if the user is located in a 'world' wherein a step forward is translated as step backwards or to the side, or if the visual manifestation of that 'world' is entirely unfamiliar to the user conceptually, then it will be difficult if not impossible to "throw" them into the VE, to locate them in a 'world' which *matters* to the user in some way. As such, not only the appearance of the 'world' itself but also the projective possibilities of that 'world'—the types of actions and interactions the 'world' *affords* to the user—must be *familiar* to the projective possibilities of objective reality.

Thus, standing in my office with my HMD in place, I confront the virtual world of *Gnomes & Goblins* (into which I have been "thrown") as a spectrum of potentialities for action (a range of possibilities onto which I project myself). This range of possibilities, then, constitutes the range of my "circumspective dealings." Driven by "curiosity," I can choose to explore the effect space, open the shutters and examine the interior of the goblins' tree-house dwellings, or pick up and play with various virtual objects within the effect space. Some of the actions I

attempt will, simply, not be available to me. I cannot, no matter how I try, kill or injure one of the small goblin's within *Gnomes & Goblins*. If I were to kill or injure the goblins, I would be unable to build the necessary relationship with the creature so that they give me the magic bell. The lack of this possibility thereby directs my attention to what is available, it literally conditions my "circumspective dealings" by allowing only certain types of actions so that I will eventually fulfill the program's ultimate operation. It is up to me to discover, through my "circumspective dealings" within the VE, what possibilities are and are not available to me. However, insofar as I act on one of these possibilities, I necessarily do not act on others; by actualizing one possibility I am choosing a way to *Be*, but the other possibilities which are not actualized remain a fundamental component of my Being for I act within the total range of possibilities. Acting within this range of possibilities works to "concernfully absorb" or harmonize the user with and within the totality of involvements which govern their actions. Because these actions are perceived by the user as the direct result of their "circumspective dealings" within the VE, the user comes to view themselves as actively implicated within the virtual world.

There is a sense, then, in which Dasein's "thrownness" and "projection"—in either a virtual world or the real one—function as a balance of determinative forces and agency. For my projective possibilities, within the concrete situation into which I am thrown, are determined by the totality of involvements. The total number of possible actions I can make within the virtual world are *always already* determined by the construction of the program's totality of involvements. As I stand in this fantastical VE the total number of actions I can execute is determined by my familiarity of interacting with and within VR programs in general, what I have been informed of by others, or what I discover by my circumspection of the dealings in which I engage with the virtual world of *Gnomes & Goblins* and its objects (although, "discover" here

does not—necessarily—imply something new overall, but something *newly* discovered for oneself); it is within this totality of involvements that has been conditioned technically and historically (the actions made available by the program itself, my total experience and familiarity with acting and interacting with and within VR programs and VEs, and what has heretofore been discovered as possible ways to *be* within a VR program in general), and culturally (the linguistic-meaning ascribed to virtual objects, and the types of actions and interactions deemed culturally acceptable)¹⁴ that I project my possibilities and actualize a select number of them. In other words, I confront the VE into which I have been “thrown” as a spectrum of potentialities for acting (a range of possibilities onto which I project myself); that range, however, is technically determined—first and foremost—by the possibilities for action afforded by the program and technology overall, and, secondly, the historically and socially conditioned ways in which *I* understand or interpret the possible, and culturally acceptable, ways to *be* in a virtual world. In summary, then, VR technology and a specific VE *technically* and *rhetorically* function to “throw” the user within the VE the totality of involvements of which is understood by the user through the historically and culturally conditioned ways in which *they* understand their Being-in-the-world of objective reality; however, it must be noted that the *rhetorical function* must operate within the bounds of the historically and culturally conditioned ways through which we understand and conceive objective reality or it will not be *familiar* and, therefore, will not successfully “throw” the user. After spending enough time being “concernfully absorbed” in this VE so that the equipment-in-use becomes phenomenologically transparent and the virtual world

¹⁴ For instance, the VR game *Dead or Alive Extreme 3: VR Paradise*, released for PlayStation VR, allows the user to “physically” molest the female characters as they verbally and physically protest. Accordingly, and rightly so, it was never released in the United States because such actions would never be deemed culturally acceptable.

and its artifacts are phenomenologically transformed to a mode of Being “ready-to-hand,” the user should be moved to Being-in-a-virtual-world-in-*the*-world.

In this mode of Being, the user uncritically accepts the phenomenological information of the virtual world and its relationship to them; this ‘world,’ in this state, is more directly and intensely “real” than their actual, and simultaneous, existence within objective reality. When a user has come to *Be*-in-a-virtual-world-in-*the*-world, they have, therefore, successfully “fallen” into that ‘world.’ If a successful Being-in-a-virtual-world-in-*the*-world can be understood as having “fallen” into such, if it characterizes how we understand that ‘world’ through “idle talk,” “curiosity,” and “ambiguity,” then such a mode of Being is “inauthentic” as it conceals any genuine understanding of that ‘world.’ Because Being-in-a-virtual-world-in-*the*-world demands that the user uncritically accept that ‘world’ as more “real” than objective reality, any resistance to this acceptance is deterred by Dasein’s or the user’s “curiosity”: their concern with the constant possibility of distraction. Ambiguity, then, manifests—when Being-in-a-virtual-world-in-*the*-world has been achieved—as the user’s inability to distinguish between their “fallen” experience and a genuine understanding of the VE: how the VE *technically* and *rhetorically* conditions their “circumspective dealings,” conditions how they *be* in the VE by manifesting itself as an ecology through its intentionally designed totality of involvements. In such a state, the user has, quite literally, turned away from their Being-in-the-world of objective reality to be absorbed into the virtual. When this “fallen” Being-in-a-virtual-world-in-*the*-world of *Gnomes & Goblins* is disrupted by the transformation, however, both the rhetorical force generated by VR technology and our “inauthentic” mode of Being-in-a-virtual-world-in-*the*-world is unconcealed because it destabilizes the fundamental structural relationship underlying any totality of involvements: the for-the-sake-of-which structure.

Let us now return to our consideration of *Gnomes & Goblins* and, particularly, the moment of disorientation produced by the transformation which occurs when one rings the bell. This transformation, as I have previously discussed, consistently produces a sense of disorientation (though nothing quite so profound as the first experience), but this consistency is conditional. One experiment I conducted while researching presence in VR was to dive into a program within which I remained for four hours. The purpose of this experiment was to test whether or not I would suffer from what is known as Alternate World Syndrome (AWS) after exiting VR and returning to objective reality. AWS is a state when in VR, as Michael Heim defines it, the user suffers a “conflict of attention” between their “biobody” and the “cyberbody.” When this happens, “an ontological rift appears as the felt world swings out of kilter,” and, subsequently, “images and expectations from an alternate world upset the current world” (210). In other words, our common understanding and expectations regarding how we understand, act, and interact in objective reality does not directly or cohesively translate into such within the virtual world, producing an “ontological rift” for the user; however, once the user has spent a sufficient amount of time within the virtual world their ontological and phenomenological orientation is recalibrated. Though Heim discusses AWS as an experience within VR, it holds true in the reverse as well. Indeed, after spending four hours within a virtual program, allowing my ontological orientation and phenomenological understanding to recalibrate, I suffered a similar disconnection, or AWS, upon my return to objective reality.¹⁵ This suggests that such technologies do indeed have the power to recalibrate the human ontological and phenomenological orientation. So, while the disorientation produced by the transformation in

¹⁵ This AWS manifested itself in the odd sensation that I remained tethered to the computer, and my ability to gage depth was slight off. This sensation lasted for around 15 minutes or so. In other words, it took around 15 minutes for my phenomenological and ontological orientation to recalibrate to my “Being-in-the-world” of objective reality.

Gnomes & Goblins is consistent, that consistency depends on the user spending enough time becoming “concernfully absorbed” into the virtual world so that they are moved to a mode of Being-in-a-virtual-world-in-*the*-world. Indeed, if one simply jumps into the program and quickly accomplishes the tasks necessary to trigger the gifting of the bell without spending a substantial amount of time within the VE, then the transformation does not produce the same sense of disorientation or at least not to the same degree. The question we must now ask is why? Why is such a disorientation produced? The “how” remains functionally the same; whether or not one spends a substantial amount of time within the VE, the transformative process is identical: the same sounds, lights, and reconfiguration of the virtual world occur regardless of the state of the user. Why, then, is the sense of disorientation which is produced dependent upon the amount of time the user spends within the VE?

“Anxiety”: An Unconcealing Event; or, a Disruption in Being-in-a-virtual-world-in-*the*-world

In “falling” Dasein “flees” in the face of itself, and, as Heidegger concludes, “[t]hat in the face of which one has “anxiety” [das Wovor der Angst] is Being-in-the-world as such” (BT 230). In light of this, it is reasonable to say that my pressing need to articulate an understanding of the disorientation produced from the transformation is prompted by “anxiety”, by my Being-in-a-virtual-world-in-*the*-world as such; for such a state of “anxiety” removes “from Dasein the possibility of understanding itself, as it falls, in terms of the ‘world’ and the way things have been publicly interpreted” (232). “Anxiety”, therefore, is a distinctive way in which the world is disclosed or opened to me: “as Dasein falls, “anxiety” brings it back from its absorption in the ‘world’. Everyday familiarity collapses, Dasein has been individualized, but individualized as Being-in-the-world. Being-in enters into the existential ‘mode’ of the “not-at-home” (233).

When I am anxious, I am “not-at-home” in the world which is to say that I am unable to find it intelligible.

Because Dasein has *always already* projected itself it always understands itself in terms of the possibilities onto which it may project; “[a]s projecting, understanding is the kind of Being of Dasein in which it is its possibilities as possibilities” (185). The transformation, in so many words, produces a sense of disorientation because it is a moment that is not understood, a moment which removes from me the possibility of understanding myself in the terms of the world of objective reality. It is a possibility which, though culturally viable as a concept, is not conceived of, consciously at least, *as* a possibility, as a way for Dasein to *Be*. This suggests, in an ontological sense, that the “anxiety” which produces the sense of disorientation derives from a unconcealing of the possibility of my “not-Being,” a possibility where Dasein *is not* “its “there”, its ‘in-the-world’” (214). The reason why the user must spend a sufficient amount of time within the VE in order to produce the sense of disorientation, therefore, is because the user must first come to *dwell* within the VE, must first come to be “at-home” in the virtual world before a mode of the “*not-at-home*” can be evoked.

“But in “anxiety” there lies the possibility of a disclosure which is quite distinctive; for “anxiety” individualizes. This individualization brings Dasein back from its falling, and makes manifest to it that authenticity and inauthenticity are possibilities of its Being” (235). Let’s say one were to come across the Boring figure (fig. 8) without knowing it was an optical illusion. Our initial



Figure 8, “An Ambiguous Image.” (Boring 445).

perception of either possibility—of either the old woman or the young woman—is not the product of critical attention and observation, but rather the product of the everyday way we perceive and understand images determined by the ‘they’. Accordingly, when one is shown that the Boring figure is an optical illusion and both images are able to be perceived simultaneously—rendering strange our initial perception of the image—a subtle disenchantment occurs: it reveals or unconceals a previously unknown possibility, an alternate way of “Being-in (and, therefore, understanding)-the-world”. The transformation in *Gnomes & Goblins* is of a similar nature, and produces an experience of my “thrownness” in the virtual world as “anxiety”, a mood which disenchant, alienates, or defamiliarizes me from my familiar Being-in-a-virtual world-in-*the-world*, from the field of everyday intelligibility determined by the ‘they’; this experience thereby discloses to me the potentiality of my not-Being through which my possibility for authentic or inauthentic Being is made aware by ontologically extracting me from the determinate force of the ‘they.’

To clarify this, let us now consider the structural relations of the totality of involvements which are manifest in the moment before the transformation is initiated from the perspective of a user who has successfully been moved to Being-in-a-virtual-world-in-*the-world*. To reiterate, in such a state, the actual equipment-in-use (the HMD and controllers) are phenomenologically transparent and the user is no longer consciously aware of their existence in objective reality; rather, their “circumspective dealings” must be understood as engaging with objects and entities within a ‘world’ in which the user exists as its “there.” I (the user) am currently working with the golden bell (a “with-which”), in the practical context of an enchanted forest (an “in-which”), in order to ring the bell and make noise (an “in-order-to”) which is directed towards fulfilling my “curiosity,” my concern for possible distraction (a “towards-this”), for-the-sake-of engaging this

‘world’ and further developing my relationship with the goblin, for-the-sake-of continuing to *Be*-in-the-virtual-world *Gnomes & Goblins*.

The moment I “ring” the bell, I am choosing to act on one of the projective possibilities afforded by this ‘world’ and the anticipated result of this act—the bell producing noise—is understood because of our “*fore-conception*”: previous knowledge of and experience with such “ready-to-hand” objects. Because every “for-the-sake-of-which” is the foundational structure of a totality of involvements, it reflects one of many potential ways for the user to *Be*. In choosing this act I am choosing a way to *Be*-in the VE, and that way is determined through a combination of the *familiar* ways in which we engage with objective reality and the projective possibilities afforded by the program. Thus, when we ring the bell and are subsequently miniaturized, this event is not understood in the moment because it is not conceived of as a projective possibility for the fact that it is an event of which we have no “*fore-having*,” “*fore-sight*,” or “*fore-conception*”: it is an event which is, simply, not available as a projective possibility in objective reality. Accordingly, this transformation functions to disrupt the “for-the-sake-of-which” structure underlying the totality of involvements thereby disrupting the *familiar* ways in which the user can *Be* and has *Been* in the VE: it moves the user from a mode of *dwelling* to a mode of “not-at-home.” Because the “for-the-sake-of-which” reflects one of our potential ways to *Be*, disrupting this structure—and, subsequently, the other structural relations which are built upon it—produces a phenomenological disorientation and ontological disruption in our orientation to the network of significances which constitute the virtual world. In other words, the user’s Being-in-a-virtual-world-in-the-world is fundamentally altered, producing the “anxiety” and ontological rift which the user experiences.

The very ability for the user to enter a mode of Being-in-*a-virtual-world-in-the-world* reveals the extent to which they are “fallen” in objective reality and alienated from a dimension or aspect of their very being: just as we cannot know a “thing-in-itself,” we cannot know ourselves in an objective sense as our very understanding of our self is primordially dependent on our “Being-in-the-world.” VR technology, therefore, does indeed function to reveal or unconceal, and that function takes the form of a phenomenological and ontological shock or disruption in our basic state of “Being-in-the-world.” To recall my earlier example of the virtual rock, our discovery of its unserviceable characteristic is a disruptive event which phenomenologically transforms the mode of Being of the virtual rock from “ready-to-hand” to “un-ready-to-hand”; this then prompts a similar phenomenological transformation of the user’s mode of Being: Dasein becomes a subject over and against a world of objects. When the power is cut to the HMD, a similar disruptive event, the phenomenological transparency of equipment-in-use is dispelled. Likewise, the miniaturization of the user is, albeit more profound, another disruptive event. By reorienting the user’s perceptual ‘world’ after having spent sufficient time becoming “at-home” within the VE, the miniaturization prompts a phenomenological transformation in the user, shifting their mode of Being from “at-home” to “not-at-home.” The disorientation produced derives from “anxiety”, from a disruption in the familiar ways in which Dasein is “Being-in-the-world.” “Anxiety,” therefore, is a disruptive event which has the potential to prompt a phenomenological transformation of Dasein’s mode of “Being-in-the-world” by making that world unintelligible. In so doing, such disruptive events work to reveal the totality of involvements which organize my “circumspective dealings” in that ‘world’ and, therefore, my ontological orientation to that ‘world.’

This unconcealing event, however, depends upon—and simultaneously reveals—the unique rhetorical force generated by virtual reality technology: its *veri(dis)similitude*. A user’s “thrownness” and “projection” in a virtual world depends on that ‘world’ being *verisimilar* to how we engage with and understand objective reality. However, though the virtual world and its mechanics may be *verisimilar* to objective reality, that is not enough to move the user to a mode of perception wherein the equipment-in-use (the HMD and controllers) become phenomenologically transparent nor does it muster enough rhetorical power to produce the phenomenological and ontological disorientation experienced.¹⁶ The transformation and subsequent “anxiety” which occurs in *Gnomes & Goblins* is only possible because it is not limited to the laws of objective reality: the fact that, as a computer program, it is not limited to earthly laws or places. Rather, its allure, its power to move the user to Being-in-a-virtual-world-in-the-world derives from that ‘world’ being *dissimilar* to our everyday experiences. The “anxiety” produced and the unconcealing function which results, depends upon the technology rhetorically inducing the user in a mode of Being-in-a-virtual-world-in-the-world which is *verisimilar* to our mode of “Being-in-the-world,” and its ability to manifest a projective possibility afforded by that ‘world’ which is entirely *dissimilar* to those available in objective reality by fundamentally altering the structural relations through which we understand our human

¹⁶ For instance, if I were to create a virtual rendition of my actual office—an everyday space with which I am intimately *familiar*—and the objects within it, chances are this would not be able to move me to a mode of Being-in-a-virtual-world-in-the-world. Even if I was able to make every object interactive, because all interactions with those objects are mediated through the handheld controllers, it is simply not possible to replicate all the possibilities of action or the sensation of resistance, say, when you pick up or move your office chair. Moreover, such spaces of everyday activity, even if translated into a VE, are so *familiar* as to not satisfy our innate “curiosity” and concern for possibilities of distraction; for if a ‘world’ is too *familiar* it will offer no encouragement for our “circumspective dealings”—the process through which we are “concernfully absorbed”—as we *already* know everything about the space and what it affords to us. Without such a conditioning process the user’s attention would be directed, rather, toward the inadequacies of the VE *in comparison* to that environment in objective reality. Therefore, in order to “concernfully absorb” a user into a VE, that VE must cater to our “curiosity,” it must offer novelty; the VE must be *dissimilar* to our everyday expectations of and experiences with objective reality.

existence in objective reality. To phrase this differently, the “anxiety” derives from a projective possibility, a way the user can *Be*, of which they have no “*fore-having*,” “*fore-sight*,” or “*fore-conception*.” The disruption, therefore, unconceals the processes through which both virtual and objective reality are made intelligible to human existence are fundamentally the same.

A virtual simulation is a human construct in which all possibilities of action are predetermined by the programmer(s); moreover, as there is always a goal or purpose, something to do or achieve, the VE is designed to condition our “circumspective dealings” in accordance with the simulation’s end design. While the constrained choice of action is illusorily perceived by the user as freedom of choice and enhances their “concernful absorption,” it also, and more importantly, constitutes the user as an object in the VE: the user is constituted as a functional object of the programmer(s) project and not something the user makes or chooses themselves to *Be*. Once we are enchanted by the spell of the virtual world, once we are moved to a mode of Being-in-a-virtual-world-in-*the*-world, we have, thus, “fallen” into the virtual world. What the transformative disruption reveals is not just another possible way for me to *Be* in a virtual world, but also the extent to which my Being-in-a-virtual-world-in-*the*-world, or any ‘world’ for that matter, is a balance of “thrownness” and “projection,” a balance of determinative forces and agency all of which—including our understanding of such—is historically and culturally conditioned. Therefore, unconcealing how the user is enchanted to a mode of Being-in-a-virtual-world-in-*the*-world through a disruption which prompts a phenomenological and ontological transformation in that mode of Being, VR functions to disenchant our “fallenness” in reality thereby revealing how we so easily succumb to the everyday patterns and practices of understanding, perceiving, and interpreting dictated by the ‘they.’

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